

FORMATTING INSTRUCTIONS FOR ICLR 2027 CONFERENCE SUBMISSIONS

Anonymous authors
Paper under double-blind review

ABSTRACT

1 INTRODUCTION

Neural networks are increasingly tasked with computations in settings that demand tight numerical tolerances, from arithmetic circuits in language models to approximating solutions of differential equations. Yet numerical fidelity remains a bottleneck for trained models (Wang & Lai, 2023; McGreivy & Hakim, 2024). For instance, Qwen2.5-7B achieves only 39.5% accuracy when computing x^2 to five decimal places (CITATION?); improving reliability through chain-of-thought or tool-calling further incurs substantial inference overhead (CITATION?). In scientific computing, small numerical errors can accumulate through repeated simulation steps, eventually corrupting long-horizon predictions (LeVeque, 2007). Learned PDE solvers still struggle to reliably attain the accuracy of established numerical methods (McGreivy & Hakim, 2024). These difficulties motivate a fundamental question: *What limits the precision attainable in neural networks, and can increasing model capacity and computation reliably yield additional fidelity?*

To isolate the **precision problem**¹, we study a simple, controlled setting: training MLPs to interpolate analytic one-dimensional functions to high precision. Although single-hidden-layer MLPs are universal function approximators (Hornik et al., 1989; 1990), gradient-based training has been documented to plateau far above machine precision even in this setting (Wang & Lai, 2023). We observe that a single-hidden-layer MLP trained to approximate x^2 plateaus 6–8 orders of magnitude above FP64 machine precision (10^{-16}). In contrast to the scaling behavior often observed elsewhere in machine learning, increasing model size or optimization compute does not reliably improve precision (Figure 1a).

Existing approximation theory likewise offers little insight into numerically realizable high-precision MLPs. Although Universal Approximation Theorems establish that sufficiently wide MLPs can approximate continuous functions arbitrarily well, known quantitative constructions exhibit a **rate–realizability tradeoff**. Classical constructions (Cybenko, 1989) achieve only algebraic convergence with width, requiring $\text{poly}(1/\epsilon)$ parameters to attain error ϵ (Figure 1b). More parameter-efficient constructions attain near-geometric rates, but incur **superlinear bit complexity** in the requested accuracy, requiring $\tilde{\Omega}(\log(1/\epsilon) \log \log(1/\epsilon))$ bits of working precision (Mhaskar,

¹Throughout, we use *precision* to refer to the numerical accuracy attainable at fixed arithmetic precision.

1996). Consequently, existing constructions provide neither a practical route to high precision nor a useful reference representation for understanding trained MLPs.

We overcome this **rate–realizability** tradeoff by constructing a practical high-precision reference solution. Building on quasi-interpolant theory (Buhmann & Jäger, 2022), we introduce QUILL (Quasi-interpolant Layers), an explicit single-hidden-layer MLP construction based on shifted localized kernels for approximating functions analytic on a neighborhood of the target domain. Unlike prior constructions, QUILL achieves geometric error convergence in width while remaining realizable in floating-point arithmetic. QUILL thereby provides both a practical route to high precision and a reference representation against which trained MLPs can be compared. Using this reference, we show that gradient descent fails to realize the width-dependent hidden-layer geometries required for high precision. Based on this insight, we instead embed QUILL directly into LLMs as a high-precision computational module as a proof-of-concept. We make the following contributions:

- **High-precision MLPs by construction:** QUILL achieves geometric error convergence with width and $O(\log(1/\varepsilon))$ parameter complexity while remaining realizable in fixed-precision arithmetic (Theorem 3.1). This matches the parameter scaling of classical spectral/polynomial approximation while using an ordinary single-hidden-layer MLP. We further show that the QUILL primitives can be composed into larger numerical computations with [XX error guarantee] and [$O(XX)$ size/depth scaling].
- **A characterization of why standard training plateaus:** QUILL identifies a critical dimensionless bandwidth parameter λ , which controls an **aliasing-conditioning** tradeoff. We show that the preferred λ -regime requires the pre-activation slopes to grow linearly with model width (Equation 8). We prove that, even with noiseless data and sufficient sampling, gradient descent fails to reach this width-dependent regime: pre-activation parameters move only polynomially slowly, while the increasingly expressive readout rapidly absorbs the residual, suppressing the gradient signal needed to further adapt the hidden-layer geometry (CITATION!).
- **High-precision QUILL modules for LLMs:** Given the optimization challenges, we instead propose to directly embed QUILL into LLMs to perform numerical computations. Because QUILL modules are ordinary feed-forward networks, they can provide high-precision numerical computation directly within pretrained models. As a proof of concept, we install constructed numerical components into frozen Gemma, Qwen, and Llama models for selected arithmetic operations. The resulting models achieve [100%] accuracy on [BENCHMARK] while using $15\times$ fewer generated tokens than chain-of-thought and $3\times$ fewer than tool calling.

2 RELATED WORK

Numerical computation in language models. Existing approaches differ largely in where numerical execution occurs. Chain-of-thought performs computation through generated intermediate reasoning steps (Wei et al., 2023); Program-of-Thoughts and PAL delegate numerical execution to an external interpreter (Chen et al., 2023; Gao et al., 2023); and tool-augmented models similarly invoke external APIs during generation (Schick et al., 2023). Recent work instead moves execution inside the model through specialized mechanisms. Dugan et al. (2024) couple a frozen LLM to a symbolic executor, while Dietz & Klakow (2025) integrate a gated calculator directly into the model. We instead encode numerical functions directly into ordinary MLP weights, allowing the numerical function to be evaluated within the network forward pass.

Arithmetic circuits and neural realizations of known computation. Arithmetic circuits represent polynomial computations as DAGs of additions and multiplications, with size and depth providing standard measures of algebraic complexity (Shpilka & Yehudayoff, 2010). Prior MLP constructions have used explicit arithmetic modules compositionally: Yarotsky (2017) construct approximate multiplication subnetworks as building blocks for smooth-function approximation. More directly, Arora et al. (2023) show that arithmetic circuits can be simulated by BaseConv gated-convolution models with controlled overhead in circuit depth and width. We use arithmetic circuits as a compilation method for composing high-precision MLP primitives into multivariate computations, with an end-to-end error bound for the resulting network (Theorem 4.1).

Constructive MLP approximation and quasi-interpolation. Universal approximation guarantees arbitrarily accurate MLPs without quantitative convergence rates (Cybenko, 1989), while constructive schemes establish explicit rates for sigmoidal networks (Jones, 1990). For analytic targets and target error ϵ , Mhaskar (1993; 1996) and De Ryck et al. (2021) obtain geometric or near-geometric convergence with width, but realizing these constructions requires working-precision bit complexity superlinear in $\log(1/\epsilon)$ (Appendix ??). Alternatively, ChebNet (Tang et al., 2019) constructs stable neural realizations of Chebyshev polynomial approximations using deep ReQU networks, with depth growing logarithmically in the polynomial degree.

Classical quasi-interpolation constructs approximants from translated localized kernels (Buhmann & Jäger, 2022). Prior work adapts quasi-interpolation to shallow MLP operators (Anastassiou, 2011; Costarelli & Spigler, 2013; Chen & Cao, 2015). More specifically, Costarelli & Spigler (2015) construct sigmoidal approximants by approximating the target derivative with localized kernels and integrating the resulting expansion. QUILL combines these constructive approximation and quasi-interpolation ideas in the high-precision regime, obtaining geometric refinement in network width while remaining realizable in fixed-precision arithmetic down to a numerical floor (Theorem 3.1).

3 CONSTRUCTING HIGH-PRECISION MLPs

Arithmetic circuits compose computations from scalar primitives; realizing these circuits in neural networks therefore requires efficient, high-precision, and numerically stable MLP primitives. Universal Approximation Theorems establish that sufficiently wide MLPs can approximate continuous functions arbitrarily well (Cybenko, 1989). However, known quantitative constructions exhibit a **rate–realizability tradeoff** (Figure 1b): classical constructions (Cybenko, 1989) achieve only algebraic convergence with width, whereas more parameter-efficient constructions attain near-geometric rates at the cost of **superlinear bit complexity** in $\log(1/\epsilon)$, where ϵ is the target error tolerance.

We resolve this tradeoff with **QUILL**: a high-precision MLP construction that achieves geometric convergence in width towards a precision floor while remaining realizable in floating-point arithmetic. Our main result, Theorem 3.1, quantifies this convergence. Theorem 3.1 also provides a bound on readout magnitudes and numerical recovery error, ensuring floating-point realizability. Finally, our analysis identifies a dimensionless bandwidth parameter λ (Equation 5) that governs the high-precision regime through an aliasing-conditioning tradeoff.

3.1 QUILL CONSTRUCTION

A quasi-interpolant viewpoint. The derivative of a tanh feature is the localized kernel

$$\frac{d}{dx} \tanh(\gamma(x - x_j)) = \gamma \operatorname{sech}^2(\gamma(x - x_j)). \quad (1)$$

QUILL exploits this identity through a quasi-interpolant framework: the kernels form a localized family of translated features whose approximation properties can be characterized in terms of their spacing and bandwidth. In particular, the theory prescribes the hidden-layer geometry and identifies the relative bandwidth required for high-precision approximation.

Construction. We approximate a scalar target f on $\Omega = [-1, 1]$. For N uniform grid cells, define

$$h = \frac{2}{N}, \quad x_j = -1 + jh, \quad j = -R, \dots, N + R. \quad (2)$$

The R nodes on either side of Ω are *halo* nodes, which provide additional degrees of freedom for correcting finite-interval boundary effects.

Choose a shared bandwidth $\gamma > 0$ and define

$$q_{\mathbf{c}}(x) = b + \sum_{j=-R}^{N+R} w_j \tanh(\gamma(x - x_j)), \quad \mathbf{c} = (b, w_{-R}, \dots, w_{N+R}). \quad (3)$$

This is a width- $W = N + 2R + 1$ one-hidden-layer tanh MLP, with input weight γ and hidden bias $-\gamma x_j$ for neuron j . Given function values at $M = sN + 1$ equispaced points $z_i \in \Omega$, where the

sampling factor s is independent of N , fitting the readout reduces to the linear least-squares problem

$$\hat{c} \in \arg \min_{c \in \mathbb{R}^{W+1}} \sum_{i=1}^M |q_c(z_i) - f(z_i)|^2. \quad (4)$$

Relative bandwidth. The quasi-interpolant geometry is governed by the center spacing h and the feature transition width γ^{-1} . Their dimensionless ratio

$$\lambda = \gamma h \quad (5)$$

measures the relative overlap of neighboring features. Theorem 3.1 shows that λ governs an aliasing–conditioning tradeoff that determines the high-precision regime, which we balance through Equation 8.

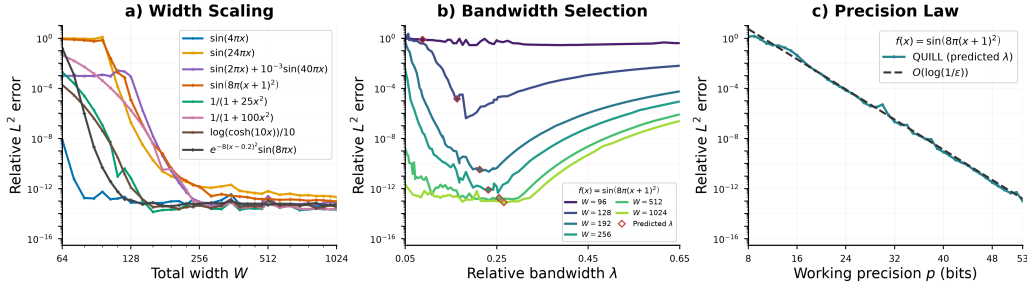


Figure 2: **QUILL achieves geometric refinement in a predictable bandwidth regime.** (a) Relative L_2 error versus width across target functions. QUILL converges geometrically with width until finite-precision recovery errors dominate. (b) Relative L_2 error versus relative bandwidth λ at several widths. The predicted bandwidth (red diamonds) (Equation 5) tracks the high-accuracy regime as the grid is refined. (c) Our QUILL construction following $O(\log(1/\epsilon))$ error scaling as it’s swept over working precision p . All solutions are computed with custom solvers restricted to p -bit precision.

3.2 HIGH-PRECISION REMAINS NUMERICALLY REALIZABLE

Theorem 3.1 controls both the continuous RMS error and the readout magnitude of the computed network.²

Theorem 3.1 (High-precision MLPs with stable readouts). *Let $f : \Omega \rightarrow \mathbb{R}$ admit a bounded analytic extension to a complex neighborhood. Fix $0 < \lambda \leq 1$, let $R = \lceil \sqrt{N} \rceil$, and choose a sufficiently large fixed observation density s . Under admissibility and numerical implementation conditions, the constructed MLP satisfies*

$$\|f - \hat{q}_W\|_{2,\Omega} \leq \underbrace{C_f e^{-c_f(\lambda)W}}_{\text{geometric refinement}} + \underbrace{\frac{C_f}{W} e^{-\pi^2/\lambda}}_{\text{aliasing}} + \underbrace{\varepsilon_{\text{rec},W}^{(2)}}_{\text{sampling and computation}}, \quad (6)$$

$$\|\bar{c}_W\|_1 \leq \underbrace{C_{\text{out}}(f, \lambda)}_{\text{width-independent readout bound}} + \underbrace{\varepsilon_{\text{coeff},W}}_{\text{coefficient formation}}. \quad (7)$$

The constants are independent of width for the fixed target class, bandwidth, and observation density; the explicit allowances, conditions, and precise theorem are given in Appendix ??.

Proof sketch. The derivative of $\tanh$ is a localized kernel, allowing quasi-interpolation to approximate the target derivative on a uniform grid. Integrating gives a $\tanh$ network, and correcting selected halo coefficients makes the boundary contribution decrease geometrically with only $O(\sqrt{N})$ halo neurons. The resulting coefficient bounds control the scaled least-squares comparison, transferring the reference network’s approximation accuracy to the computed readout, with the sampling

²We write $\|g\|_{2,\Omega} := (\frac{1}{2} \int_{-1}^1 |g(x)|^2 dx)^{1/2}$.

Construction	Maximum Width	Largest Weight	Working Precision (bits)	Hidden Layers
Classical staircase [†]	$O(\varepsilon^{-1})$	$O(\varepsilon^{-1})^\dagger$	$O(\log(1/\varepsilon))^\dagger$	1
Mhaskar (1996)	$O(\log(1/\varepsilon))$	$\exp[\tilde{O}(\log^2(1/\varepsilon))]^\dagger$	$\tilde{O}(\log^2(1/\varepsilon))^\dagger$	1
Costarelli–Spigler (2015)	$O(\varepsilon^{-1})^\dagger$	$O(\varepsilon^{-1})^\dagger$	$O(\log(1/\varepsilon))^\dagger$	1
ChebNet (2019)	$O(\log(1/\varepsilon))$	$O(1)^\dagger$	$O(\log(1/\varepsilon))^\dagger$	$O(\log \log(1/\varepsilon))$
QUILL	$O(\log(1/\varepsilon))$	$O(\log(1/\varepsilon))$	$O(\log(1/\varepsilon))$	1

Table 1: Rates for a fixed univariate analytic class at error ε . Precision covers construction and evaluation. All rates are sufficient, not necessary. $\tilde{O}$ hides powers of $\log \log(1/\varepsilon)$.

and arithmetic allowances shown above. The analytic reference coefficients are used only in the proof; full arguments appear in Appendices ??, ??, and ??.

Theorem 3.1 implies $O(\log(1/\varepsilon))$ width for target error ε .

Corollary 3.1.1 (Logarithmic-width high-precision MLPs). *Under the conditions of Theorem 3.1, let ε be a target tolerance above the finite-precision floor. Choosing λ to make the aliasing contribution sufficiently small, QUILL attains*

$$\|f - \hat{q}_W\|_{2,\Omega} \leq \varepsilon, \quad W = O\left(\log \frac{1}{\varepsilon}\right).$$

Section 3.3 gives a practical rule for selecting λ from the sampled target and requested tolerance.

Table ?? compares QUILL against representative classical MLP constructions. Together, Theorem 3.1 and Corollary 3.1.1 resolve the rate–realizability tradeoff: QUILL achieves logarithmic width in the target accuracy without the superlinear working precision or polynomially growing parameter magnitudes of prior parameter-efficient constructions. The largest-parameter column includes the hidden weights and biases, whereas Theorem 3.1 separately controls the total readout magnitude.

Finite precision floor. The term $\epsilon_{\text{rec},W}$ accounts explicitly for the least-squares solve, coefficient formation, and floating-point evaluation. Under the implementation assumptions of Theorem 3.1 these terms remain controlled, so the approximation error decreases geometrically until floating-point recovery dominates. Figure 2(a,c) verifies this behavior: increasing working precision lowers the observed floor. Detailed roundoff bounds are given in Appendix XX.

3.3 HIGH-PRECISION BANDWIDTHS ARE PREDICTABLE

Rather than tuning λ by hyperparameter search, we predict it from the sampled target and an effective aliasing tolerance. Smaller λ suppresses aliasing but increases the bounds on readout amplification and boundary-correction cost. Matching the leading aliasing score to an effective tolerance ε_{eff} :

$$2 \exp\left(-\frac{\pi^2}{\lambda_{\text{pred}}}\right) \sinh\left(\frac{2\pi\bar{\omega}}{N\lambda_{\text{pred}}}\right) = \varepsilon_{\text{eff}}, \quad (8)$$

where N is the number of interior grid cells and $\bar{\omega}$ is the amplitude-weighted mean absolute angular frequency, estimated from the sampled target’s Fourier coefficients. Figure 2(b) shows that λ_{pred} tracks the observed high-precision bandwidth regime. Appendix ?? gives the derivation and frequency-estimation details.

3.4 BANDWIDTH CONTROLS OPTIMIZATION ACCESS TO HIGH PRECISION

The construction’s nearly width-independent relative bandwidth $\lambda = \gamma h$ requires $\gamma = \Theta(W)$, yet trained RMS slopes fall short (Figure ??). Even though larger widths admit smaller λ at comparable

[†]Derived for the implementations in Appendix ??; not lower bounds.

[‡]Staircase specified there; Jones (1990) is the historical reference, not the source of these rates.

construction error (Figure 2b), small slopes can delay learning by weakening target-relevant kernel directions. We quantify this delay by replacing Section 3.1’s direct least-squares solve with GD. This connects prior work on spectral bias (??) to readout optimization in the quasi-interpolant dictionary.

Theorem 3.2 (Slope-dependent output-error lower bound). *For the fixed-center construction of Section 3.1 with common slope $\gamma > 0$, let $\mu_1 \geq \mu_2 \geq \dots \geq 0$ be the normalized readout-kernel eigenvalues. For a nonzero sampled target, readout GD from zero with $\eta = 1/(2\mu_1)$ satisfies*

$$\frac{\|f - q_n\|_{2,Z}^2}{\|f\|_{2,Z}^2} \geq \sum_i p_i(\gamma) \left(1 - \frac{\bar{p}_i(\gamma)}{2}\right)^{2n}, \quad n \geq 0. \quad (9)$$

Here $\|\cdot\|_{2,Z}$ is the training-sample RMS norm, p_i is the fraction of sampled target energy projected onto kernel eigenvector i , and $\mu_i/\mu_1 \leq \bar{p}_i(\gamma) \leq 1$. The explicit gamma-dependent rate bounds and finite-geometry allowances are given in Appendix ??.

Proof sketch. The readout kernel contains normalized inner products of feature vectors at pairs of training points. To bound its spectrum, we separate the constant sample direction and approximate the sum over centers by an integral. This approximation depends explicitly on $d \coth(\gamma d)$, where d separates those points. Bounding the finite-center corrections gives $\bar{p}_i(\gamma)$; standard GD eigenmode decay then yields the output-error bound.

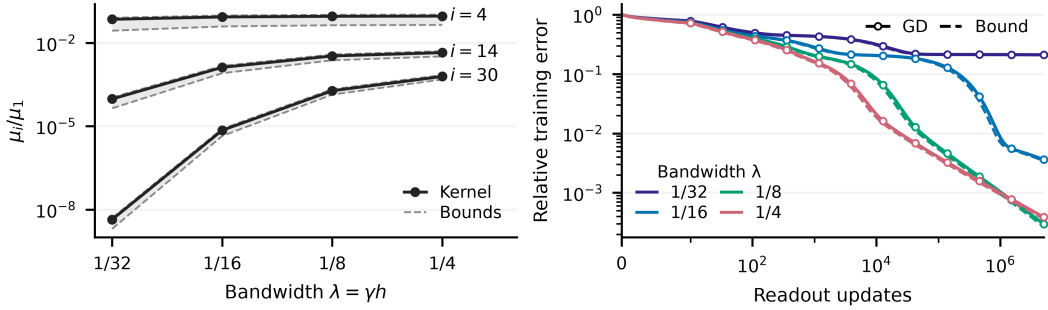


Figure 3: **Small slopes weaken kernel directions and delay readout learning.** Width $W = 512$; target $f(x) = [\sin(2\pi x) + \frac{1}{2}\sin(6\pi x) + \frac{1}{4}\sin(14\pi x)]/\sqrt{21/32}$ on $[-1, 1]$. **Left:** Kernel ratios μ_i/μ_1 (dots) and intervals from Theorem ?? (dashed boundaries, shading) for three ranks carrying substantial target energy. Lines connect ordered ranks; eigenvectors change with bandwidth. **Right:** Executed frozen-readout GD errors (solid, hollow markers) and square roots of Theorem 3.2’s lower bounds (dashed). The error bound combines theorem-derived rate bounds with all resolved target projections onto the frozen kernel’s eigenvectors. Errors are relative training RMS through five million updates; geometry and protocol are in Appendix ??.

The spectral bounds of Theorem ?? track how bandwidth changes target-relevant ratios μ_i/μ_1 (Figure 4, left). GD’s step size is limited by the largest eigenvalue μ_1 . At small bandwidths, target-relevant eigenvalues are much smaller, so the corresponding error components decay slowly. Theorem 3.2 converts these rate bounds into output-error lower bounds that closely track executed GD (Figure 4, right): small slopes delay accuracy already available in the dictionary.

Joint training shows an empirical acquisition shortfall under both GD and Adam: the population’s mean bandwidth remains below the supplied construction scale, with a substantial output-error gap over the same training budget (Figure ??). We next examine why training fails to acquire useful γ .

3.5 TRAINING FAILS TO SELF-REINFORCING SLOPE ACQUISITION

4 GENERALIZATION VIA ARITHMETIC CIRCUITS

The preceding results show that bounded analytic scalar functions can be approximated to high precision in MLP weights. Here we show that these scalar constructions can be composed into

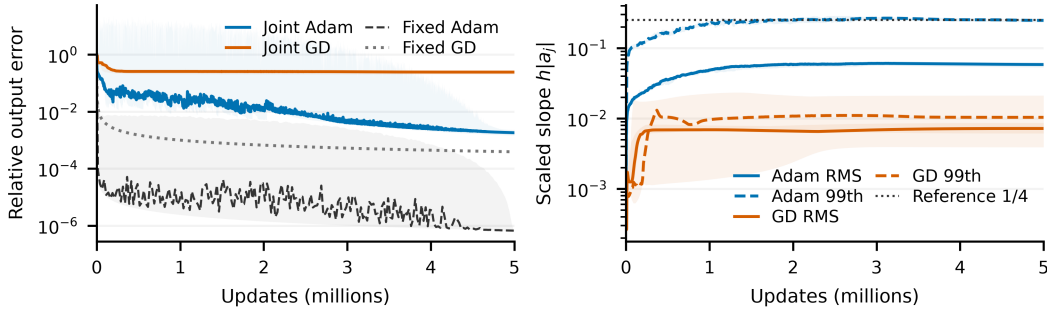


Figure 4: Needs to be updated to a 3-panel figure

Table 2: **QUILL primitives compose into broader numerical and discrete programs.** maybe i should move this to the end of section and do banner fig instead? S_{QI} counts executed scalar QUILL modules, and D_{QI} is the longest sequential dependency path through those modules; affine wiring is not counted. Numerical rows report empirical held-out FP64 approximation or implementation error on the stated bounded domains, while discrete rows report accuracy after threshold decoding. Newton and quadrature report implementation error relative to the exact supplied finite algorithm rather than total error to the limiting mathematical target.

Class	Program	Input size	S_{QI}	D_{QI}	QUILL primitives	Empirical FP64 result
Polynomial	Dot product ($d = 4$)	8 scalars	8	1	square	rel. L^2 : 8.00×10^{-15}
Polynomial	3×3 determinant	9 scalars	24	2	square	rel. L^2 : 1.55×10^{-14}
Analytic	3D Gaussian RBF	6 scalars	4	2	square, exp	rel. L^2 : 5.28×10^{-14}
Analytic	3-logit softmax	3 logits	11	3	square, exp, recip., log	rel. L^2 : 4.36×10^{-14}
Algorithm	Newton reciprocal, 5 steps	1 scalar	20	10	square	impl. max: 3.11×10^{-14}
Algorithm	8-node quadrature	1 frequency	40	3	square, exp, cos	impl. max: 4.42×10^{-14}
Discrete	Byte equality to A	8 bits	2	2	equality, restoration	100% decode
Discrete	Fixed 4-clique, 8 vertices	28 edge bits	71	2	fan-in gates	100% decode

much richer computations. Compiling supplied arithmetic circuits with QUILL modules yields neural realizations of bounded multivariate polynomials, including quadratic forms, matrix products, and determinants; additional scalar modules extend the framework to non-polynomial functions and fixed-step numerical algorithms. The same approach also supports discrete operations such as equality checks, pattern matching, and bounded Boolean circuits.

As a proof of concept, we demonstrate internal tool calling: we insert a constructed x^2 module into frozen pretrained language models and execute it during the forward pass, showing that these programmed computations can run inside existing neural architectures rather than through token-by-token reasoning or an external tool.

4.1 FROM ARITHMETIC CIRCUITS TO NEURAL PROGRAMS

Arithmetic circuits provide a standard way to build multivariate polynomial computations from addition and multiplication. An arithmetic circuit is a directed acyclic graph whose inputs are variables or constants and whose internal nodes, called *gates*, perform these operations. Intermediate values may be reused, and independent branches may be evaluated in parallel. Many familiar computations, including polynomial evaluation, matrix products, quadratic forms, and determinants, admit compact arithmetic circuits (?).

Compiling arithmetic circuits. QUILL realizes these circuits directly in neural weights. Addition, subtraction, and multiplication by constants are affine operations implemented directly by neural weights. Multiplication is the only nonlinear primitive and follows from the scalar x^2 QUILL

through polarization:

$$m_\eta(x, y) = xy = \left(\frac{x+y}{2}\right)^2 - \left(\frac{x-y}{2}\right)^2, \quad |\tilde{m}_\eta(x, y) - xy| \leq 2\eta, \quad (10)$$

Thus, if $\tilde{q}_\eta$ approximates t^2 with uniform error at most η on the required interval, the induced neural multiplier xy has error at most 2η . (cite pf in appendix)

Because affine gates require no constructed nonlinear module, we characterize the compilation using the number of multiplication gates $S_\times(\Phi)$ and the multiplicative depth $D_\times(\Phi)$, defined as the maximum number of multiplication gates along any input–output path. These quantities determine the hidden units and sequential nonlinear depth of the compiled program; the numerical accuracy additionally depends on how the supplied circuit amplifies local gate errors.

Error bounds. Let $N_{\Phi, \delta}$ denote the neural network obtained by replacing each multiplication gate of Φ with a QUILL multiplier of uniform error at most δ . Let $S_\times(\Phi)$ be the number of multiplication gates and $D_\times(\Phi)$ the maximum number along any input–output path.

Assume all intermediate values of the exact and compiled circuits remain in fixed bounded intervals. Propagating local errors through the circuit gives

$$\|N_{\Phi, \delta} - f_\Phi\|_{\infty, \Omega} \leq \kappa_\Phi \delta,$$

where κ_Φ is the circuit’s error-amplification factor. It is obtained by composing the local sensitivities of successive gates and therefore depends on both their numerical conditioning and the circuit depth; in the worst case it may grow exponentially with $D_\times(\Phi)$.

Theorem 4.1 (Compilation of bounded arithmetic circuits). *If each QUILL multiplier uses $w(\delta)$ nonlinear units and depth $\ell(\delta)$, then the compiled network uses*

$$O(S_\times(\Phi)w(\delta)) \text{ nonlinear units,} \quad O(D_\times(\Phi)\ell(\delta)) \text{ nonlinear depth.}$$

To achieve final error ε , it suffices to take $\delta \leq \varepsilon/\kappa_\Phi$, provided this accuracy is above the finite-precision floor.

In the geometric-refinement regime, $w(\delta) = O(\log(1/\delta))$ and $\ell(\delta) = O(1)$, giving

$$W_{\text{nl}} = O\left(S_\times(\Phi) \left[1 + \log_+ \frac{\kappa_\Phi}{\varepsilon}\right]\right), \quad D_{\text{nl}} = O(D_\times(\Phi)).$$

Proof sketch. Let E_v denote the propagated error at gate v . Affine gates satisfy $E_v \leq \sum_i |a_i| E_{u_i}$. For a multiplication gate $v = uw$,

$$E_v \leq B_w E_u + B_u E_w + E_u E_w + \delta,$$

where B_u and B_w bound the exact and approximate values entering the gate. Induction through the DAG gives an output bound $E_\Phi(\delta)$ that is a polynomial in δ with nonnegative coefficients and zero constant term. Thus $E_\Phi(\delta)/\delta$ is nondecreasing, and with $\kappa_\Phi := E_\Phi(\delta_{\max})/\delta_{\max}$, $E_\Phi(\delta) \leq \kappa_\Phi \delta$ for $0 < \delta \leq \delta_{\max}$. The size and depth bounds follow gatewise. Full details are in Appendix ??.

Corollary 4.1.1 (QUILL arithmetic circuits). *In the geometric-refinement regime, $w(\delta) = O(\log(1/\delta))$ and $\ell(\delta) = O(1)$. Hence, for fixed circuit domains,*

$$W_{\text{nl}} = O\left(S_\times(\Phi) \left[1 + \log_+ \left(\frac{\kappa_\Phi}{\varepsilon}\right)\right]\right), \quad D_{\text{nl}} = O(D_\times(\Phi)), \quad (11)$$

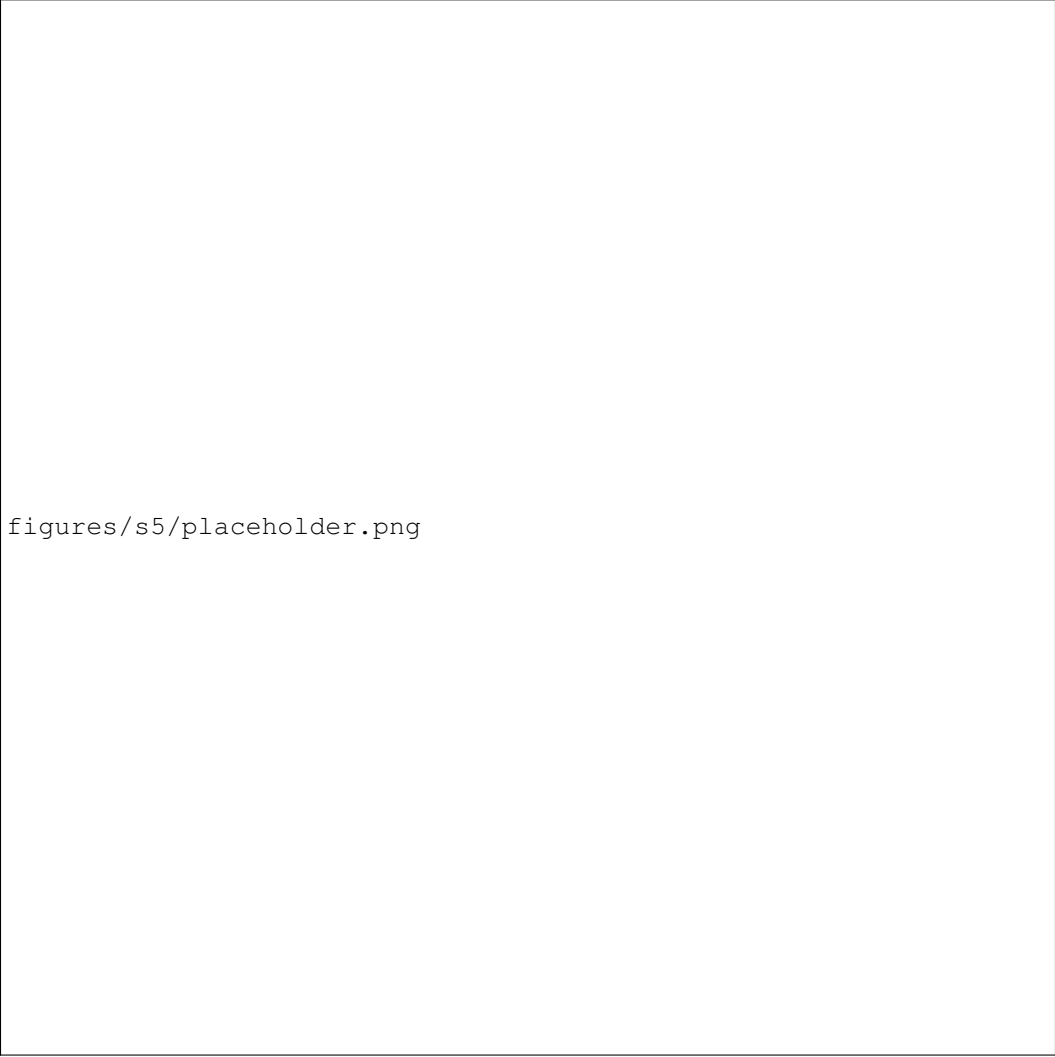
up to the finite-precision floor, where $\log_+(x) := \max\{0, \log x\}$.

Multivariate polynomials from arithmetic circuits. The circuit theorem immediately extends the scalar QUILL construction to multivariate polynomial computation. For a polynomial with M monomials and total degree at most d , balanced multiplication trees give

$$S_\times = O(Md), \quad D_\times = O(\log d),$$

with smaller circuits possible when algebraic structure is reused.

We evaluate this construction on multiplication, dot products, squared distances, bilinear and quadratic forms, covariance, multilinear polynomials, tensor contractions, and determinants. Across this suite, the largest sampled relative L_2 error is 6.46×10^{-14} and the largest sampled absolute error is 5.04×10^{-13} . Figure 5 compares these compiled QUILL programs with shallow multivariate models fit directly from samples.



figures/s5/placeholder.png

Figure 5: **PLACEHOLDER** DAG visual / pipeline?

4.2 BEYOND POLYNOMIAL ARITHMETIC.

Arithmetic circuits contain only addition and multiplication, so they compute polynomials. We extend the available primitives with directly constructed one-dimensional QUILL modules for bounded functions such as reciprocal, exponential, sine, and inverse square root. These modules compose with the arithmetic compiler to represent functions such as Gaussian RBFs, softmax, cosine similarity, and bounded matrix inversion.

Adding non-polynomial primitives The same composition and error-propagation framework applies, provided every exact and approximate intermediate value remains within the domain prescribed for its corresponding scalar module. For an additional unary gate $v = g(u)$, suppose g is L_v -Lipschitz on its prescribed domain and its constructed realization has uniform local error at most δ_v . Then

$$E_v \leq L_v E_u + \delta_v.$$

Thus the same gatewise analysis extends from arithmetic circuits to a supplied finite library of bounded unary functions, with the corresponding local errors and sensitivities included in the propagated bound.

The representation is compact because the target is supplied as a composition of simple low-dimensional operations, not merely because the full multivariate function is analytic. Across the tested augmented programs, the maximum sampled FP64 implementation errors are 9.94×10^{-14} for a three-dimensional Gaussian RBF, 1.46×10^{-13} for three-logit softmax, and 1.79×10^{-13} for three-dimensional cosine similarity.

Finite algorithms. A fixed number of algorithmic steps can likewise be unrolled into a finite graph. For an algorithm A , its QUILL implementation $\hat{A}$, and target f ,

$$|\hat{A} - f| \leq \underbrace{|\hat{A} - A|}_{\text{implementation}} + \underbrace{|A - f|}_{\text{algorithmic}}.$$

QUILL controls the first term; the second is inherited from the source algorithm. For reciprocal Newton iteration, four compiled steps agree with the exact four-step algorithm to 3.02×10^{-14} , even though the algorithm itself remains 3.10×10^{-10} from $1/x$; after five steps, total error reaches 3.06×10^{-14} . An eight-node Gaussian quadrature example similarly has 4.42×10^{-14} implementation error while its quadrature error remains 5.84×10^{-8} .

Boolean circuits. The same construction extends to discrete computation. For Boolean inputs $x, y \in \{0, 1\}$,

$$x \wedge y = xy, \quad x \vee y = x + y - xy, \quad \neg x = 1 - x,$$

so affine operations and the QUILL multiplier already form a complete bounded-fan-in Boolean gate set. Because approximate gates may drift away from $\{0, 1\}$, we optionally apply a restoring scalar map that contracts perturbations around the two Boolean states. This yields stable composition of deeper Boolean circuits

Across AND, OR, majority, and k -of- n gates, all tested configurations achieve 100% accuracy on legal Boolean inputs. Random Boolean circuits decode correctly through depth 32, and fixed substring checks achieve 100% accuracy. Composing the same primitives with standard circuit structures yields larger discrete programs: bitonic sorting is exact for up to 32 eight-bit integers, and the min-plus shortest-path construction returns correct rounded distances through $n = 16$.

We treat this as a discrete compilation result rather than a classical arithmetic-circuit theorem: the source circuit is supplied, correctness is defined after Boolean decoding on legal or perturbed states, and no asymptotic AC^0 or P/poly simulation claim is made.

Discrete algorithms. Composing these primitives with standard circuit structures yields larger discrete programs. Bitonic sorting achieved 100% exact-array accuracy for up to 32 eight-bit integers in both FP32 and FP64, with or without restoration. A min-plus shortest-path construction achieved 100% rounded-distance accuracy and valid decoded predecessor choices through $n = 16$, with worst raw distance error 0.188.

Representation limits and extensions. The compiled program inherits the limits of its numerical representation. For example, FP64 cannot represent every integer above 2^{53} exactly, so exact large-integer arithmetic requires a multiword representation such as digits or limbs; algorithms such as Karatsuba can then be compiled over those primitives. Likewise, a fixed feed-forward network realizes only a fixed number of algorithmic steps, while data-dependent or unbounded iteration requires repeated execution or an external controller. Finally, gate-by-gate composition need not be optimal: a small subgraph may instead be replaced by a single validated QUILL module. Thus representation, decomposition, and fusion jointly determine the size and numerical stability of the resulting neural program.

5 PROOF-OF-CONCEPT LLM INSTALLATION

The preceding results show that known numerical and discrete computations can be encoded directly in neural weights. As a proof of concept, we test whether the same constructed weights can execute inside an existing pretrained model.

Digits	Model	Direct	CoT	Calculator	Frozen QI
3	Gemma-2-2B	38.3% · 9.0	12.5% · 104.4	100% · 22.7	100% · 5.7
	Qwen2.5-7B	77.3% · 6.6	3.1% · 189.5	100% · 19.7	100% · 5.7
	Llama-3.1-8B	24.2% · 3.4	24.2% · 166.0	100% · 12.0	100% · 5.7
5	Gemma-2-2B	7.8% · 11.7	3.1% · 144.6	100% · 30.7	100% · 9.7
	Qwen2.5-7B	12.5% · 10.1	5.5% · 179.6	100% · 27.7	100% · 9.7
	Llama-3.1-8B	3.1% · 5.2	2.3% · 187.9	100% · 15.7	100% · 9.7
7	Gemma-2-2B	0% · 16.4	1.6% · 183.5	95.3% · 38.6	100% · 13.7
	Qwen2.5-7B	9.4% · 12.7	0% · 168.5	100% · 35.7	100% · 13.7
	Llama-3.1-8B	0.8% · 7.4	0% · 190.5	100% · 19.0	100% · 13.7
Average		19.3% · 9.2	5.8% · 168.3	99.5% · 24.6	100% · 9.7

Table 3: **End-to-end integer-squaring results.** Each model–digit condition contains $n = 128$ examples (1,152 evaluations total). Each entry reports exact accuracy · mean generated output tokens; accuracies are rounded to one decimal place. Frozen QI routes the supplied operand through a constructed width-80 scalar MLP during the neural forward pass.

5.1 INSTALLATION

Transformer language models contain MLP blocks, so a constructed QUILL module can be inserted directly into the model’s forward computation. We install a frozen width-80 x^2 QUILL into Gemma-2-2B, Qwen2.5-7B, and Llama-3.1-8B while keeping all pretrained weights fixed.

Our pathway has three steps: it reads a supplied operand from the residual stream, evaluates x^2 with the inserted QUILL module, and writes the serialized result back for decoding by the frozen LM head. Triggering, operand extraction, and serialization are hard-coded in this demonstration.

5.2 RESULTS

We evaluate 3-, 5-, and 7-digit inputs in Gemma-2-2B, Qwen2.5-7B, and Llama-3.1-8B against direct generation, chain of thought, and calculator use. Frozen QI executes x^2 during the forward pass and generates only the answer.

Across all nine model–digit settings, Frozen QI achieves 100% exact accuracy with 9.7 generated tokens on average, compared with 99.5% and 24.6 tokens for calculator use and 5.8% and 168.3 tokens for chain of thought. This experiment establishes that a constructed QUILL function can execute inside the forward pass of an otherwise frozen pretrained model.

(swap note: not arguing not that an MLP is a better calculator than conventional arithmetic software. Rather, this experiment shows that a supplied numerical capability can be encoded directly in ordinary neural-network weights and executed inside a frozen pretrained model. When a computation is known, frequently used, and small enough to reside inside the model, internal execution can avoid the interface overhead of reconstructing it token by token or invoking an external tool.)

6 DISCUSSION

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, “we checked LLM-generated research ideas for potential plagiarism through a manual literature survey”, “LLM-generated code was verified and tested for correctness by 2 authors”, etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

See the ICLR 2027 AI Policy for Authors for more details. This statement should not be more than 1 page.

ETHICS STATEMENT

(This section is **recommended** and does not count toward the page limit.)

If authors feel that their paper submission raises questions regarding the Code of Ethics, they are encouraged to include a paragraph of Ethics Statement (at the end of the main text before references) to address potential concerns where appropriate. Topics include, but are not limited to, studies that involve human subjects, practices to data set releases, potentially harmful insights, methodologies and applications, potential conflicts of interest and sponsorship, discrimination/bias/fairness concerns, privacy and security issues, legal compliance, and research integrity issues (e.g., IRB, documentation, research ethics). This statement should not be more than 1 page.

REPRODUCIBILITY STATEMENT

(This section is **recommended** and does not count toward the page limit.)

It is important that the work published in ICLR is reproducible. Authors are strongly encouraged to include a paragraph-long Reproducibility Statement at the end of the main text (before references) to discuss the efforts that have been made to ensure reproducibility. This paragraph should not itself describe details needed for reproducing the results, but rather reference the parts of the main paper, appendix, and supplemental materials that will help with reproducibility. For example, for novel models or algorithms, a link to an anonymous downloadable source code can be submitted as supplementary materials; for theoretical results, clear explanations of any assumptions and a complete proof of the claims can be included in the appendix; for any datasets used in the experiments, a complete description of the data processing steps can be provided in the supplementary materials. Each of the above are examples of things that can be referenced in the reproducibility statement.

AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

REFERENCES

- Ben Adcock and Alexei Shadrin. Fast and stable approximation of analytic functions from equispaced samples via polynomial frames, 2022. URL <https://arxiv.org/abs/2110.03755>.
- George A. Anastassiou. Univariate hyperbolic tangent neural network approximation. *Mathematical and Computer Modelling*, 53(5):1111–1132, 2011. ISSN 0895-7177. doi: <https://doi.org/10.1016/j.mcm.2010.11.072>. URL <https://www.sciencedirect.com/science/article/pii/S0895717710005649>.
- Edward Anderson, Zhaojun Bai, Christian Bischof, L. Susan Blackford, James Demmel, Jack Dongarra, Jeremy Du Croz, Anne Greenbaum, Sven Hammarling, Alan McKenney, and Danny Sorensen. *LAPACK Users’ Guide*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 3 edition, 1999. doi: 10.1137/1.9780898719604. URL <https://doi.org/10.1137/1.9780898719604>.
- Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. Zoology: Measuring and improving recall in efficient language models, 2023. URL <https://arxiv.org/abs/2312.04927>.
- Claude Brezinski and Michela Redivo-Zaglia. New representations of padé, padé-type, and partial padé approximants. *Journal of Computational and Applied Mathematics*, 284:69–77, 2015. doi: 10.1016/j.cam.2014.07.007. URL <https://doi.org/10.1016/j.cam.2014.07.007>.
- Martin Buhmann and Janin Jäger. *Quasi-Interpolation*. Cambridge Monographs on Applied and Computational Mathematics. Cambridge University Press, 2022.
- Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks, 2023. URL <https://arxiv.org/abs/2211.12588>.
- Zhi-xiang Chen and Fei-long Cao. The construction and approximation of feedforward neural network with hyperbolic tangent function. *Applied Mathematics-A Journal of Chinese Universities*, 30(2):151–162, 2015. doi: 10.1007/s11766-015-3000-9. URL <https://doi.org/10.1007/s11766-015-3000-9>.
- D. Costarelli and R. Spigler. Constructive approximation by superposition of sigmoidal functions. *Analysis in Theory and Applications*, 29(2):169–196, 2013. doi: 10.4208/ata.2013.v29.n2.8. URL <https://doi.org/10.4208/ata.2013.v29.n2.8>.
- Danilo Costarelli and Renato Spigler. Approximation by series of sigmoidal functions with applications to neural networks. *Annali di Matematica Pura ed Applicata (1923 -)*, 194(1):289–306, 2015. doi: 10.1007/s10231-013-0378-y. URL <https://doi.org/10.1007/s10231-013-0378-y>.
- George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4):303–314, 1989. doi: 10.1007/BF02551274.
- Tim De Ryck, Samuel Lanthaler, and Siddhartha Mishra. On the approximation of functions by tanh neural networks. *Neural Networks*, 143:732–750, November 2021. ISSN 0893-6080. doi: 10.1016/j.neunet.2021.08.015. URL <http://dx.doi.org/10.1016/j.neunet.2021.08.015>.
- Florian Dietz and Dietrich Klakow. Igc: Integrating a gated calculator into an llm to solve arithmetic tasks reliably and efficiently, 2025. URL <https://arxiv.org/abs/2501.00684>.
- Owen Dugan, Donato Manuel Jimenez Beneto, Charlotte Loh, Zhuo Chen, Rumen Dangovski, and Marin Soljačić. Occamllm: Fast and exact language model arithmetic in a single step, 2024. URL <https://arxiv.org/abs/2406.06576>.
- Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. Pal: Program-aided language models, 2023. URL <https://arxiv.org/abs/2211.10435>.

-
- Walter Gautschi. On inverses of vandermonde and confluent vandermonde matrices. *Numerische Mathematik*, 4:117–123, 1962. doi: 10.1007/BF01386302. URL <https://doi.org/10.1007/BF01386302>.
- Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2 edition, 2002. doi: 10.1137/1.9780898718027. URL <https://doi.org/10.1137/1.9780898718027>.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989. ISSN 0893-6080. doi: [https://doi.org/10.1016/0893-6080\(89\)90020-8](https://doi.org/10.1016/0893-6080(89)90020-8). URL <https://www.sciencedirect.com/science/article/pii/0893608089900208>.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Universal approximation of an unknown mapping and its derivatives using multilayer feedforward networks. *Neural Networks*, 3(5):551–560, 1990. ISSN 0893-6080. doi: [https://doi.org/10.1016/0893-6080\(90\)90005-6](https://doi.org/10.1016/0893-6080(90)90005-6). URL <https://www.sciencedirect.com/science/article/pii/0893608090900056>.
- Mohsin Javed and Lloyd N. Trefethen. A trapezoidal rule error bound unifying the euler–maclaurin formula and geometric convergence for periodic functions. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 470(2161):20130571, 2014. doi: 10.1098/rspa.2013.0571. URL <https://doi.org/10.1098/rspa.2013.0571>.
- L.K. Jones. Constructive approximations for neural networks by sigmoidal functions. *Proceedings of the IEEE*, 78(10):1586–1589, 1990. doi: 10.1109/5.58342.
- Randall J. LeVeque. *Finite Difference Methods for Ordinary and Partial Differential Equations*. Society for Industrial and Applied Mathematics, 2007. doi: 10.1137/1.9780898717839. URL <https://epubs.siam.org/doi/abs/10.1137/1.9780898717839>.
- Nick McGreivy and Ammar Hakim. Weak baselines and reporting biases lead to overoptimism in machine learning for fluid-related partial differential equations. *Nature Machine Intelligence*, 6(10):1256–1269, September 2024. ISSN 2522-5839. doi: 10.1038/s42256-024-00897-5. URL <http://dx.doi.org/10.1038/s42256-024-00897-5>.
- H. N. Mhaskar. Neural networks for optimal approximation of smooth and analytic functions. *Neural Computation*, 8(1):164–177, 1996. doi: 10.1162/neco.1996.8.1.164.
- Hrushikesh Narhar Mhaskar. Approximation properties of a multilayered feedforward artificial neural network. *Advances in Computational Mathematics*, 1(1):61–80, 1993.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools, 2023. URL <https://arxiv.org/abs/2302.04761>.
- Amir Shpilka and Amir Yehudayoff. Arithmetic circuits: A survey of recent results and open questions. *Found. Trends Theor. Comput. Sci.*, 5:207–388, 2010. URL <https://api.semanticscholar.org/CorpusID:86398017>.
- Shanshan Tang, Bo Li, and Haijun Yu. ChebNet: Efficient and stable constructions of deep neural networks with rectified power units via chebyshev approximations, 2019. URL <https://arxiv.org/abs/1911.05467>.
- Lloyd N. Trefethen. *Approximation Theory and Approximation Practice, Extended Edition*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2019. doi: 10.1137/1.9781611975949. URL <https://epubs.siam.org/doi/abs/10.1137/1.9781611975949>.
- Yongji Wang and Ching-Yao Lai. Multi-stage neural networks: Function approximator of machine precision, 2023. URL <https://arxiv.org/abs/2307.08934>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models, 2023. URL <https://arxiv.org/abs/2201.11903>.
- Dmitry Yarotsky. Error bounds for approximations with deep relu networks, 2017. URL <https://arxiv.org/abs/1610.01145>.

A QUILL CONSTRUCTION THEORY

This section proves the construction and numerical-realizability results used in Sections 3.1–3.2. The proof follows four stages: construct an accurate tanh reference network on the prescribed grid; correct its finite-interval boundary contribution while controlling the readout; recover a comparably accurate readout from in-domain samples by reference-scaled truncated SVD; and transfer the sampled residual to continuous RMS with an explicit finite-precision allowance.

A.1 SETUP AND NOTATION

A.1.1 TARGET CLASS AND PRESCRIBED TANH DICTIONARY

Let

$$\Omega = [-1, 1], \quad I_\Delta = [-1 - \Delta, 1 + \Delta],$$

and, for $\delta, B > 0$, let $\mathcal{A}_{\delta, B}(I_\Delta)$ denote the functions that are real-valued on I_Δ , holomorphic on

$$U_\delta(I_\Delta) = \{z \in \mathbb{C} : \text{dist}(z, I_\Delta) < \delta\},$$

and bounded there by B .

Fix $0 < \lambda \leq 1$ and a halo-design constant $a_H > 0$. For an integer $N \geq 1$, set

$$h = \frac{2}{N}, \quad x_j = -1 + jh, \quad R = \lceil a_H \sqrt{N} \rceil, \quad \gamma = \frac{\lambda}{h}, \quad W = N + 2R + 1. \quad (\text{A.1})$$

The prescribed width- W tanh dictionary is

$$q_c(x) = b + \sum_{j=-R}^{N+R} w_j \tanh(\gamma(x - x_j)), \quad \|c\|_1 = |b| + \sum_j |w_j|. \quad (\text{A.2})$$

A width is called *admissible* if

$$\left(R + \frac{1}{2}\right)h \leq \Delta, \quad \gamma\delta \geq 4\pi, \quad \lambda R \geq 2 \log 2, \quad N \geq 16 \max\{1, a_H^2\}. \quad (\text{A.3})$$

The derivative-kernel viewpoint behind equation A.2 is classical in constructive sigmoidal approximation: one may approximate a derivative using a localized density and integrate the expansion to obtain a sigmoidal network. See, for example, Costarelli & Spigler (2015) and, for broader quasi-interpolation background Buhmann & Jäger (2022).

A.1.2 SAMPLES AND NUMERICAL OUTPUT

Let $s \geq 2$ be an integer and sample only on the core interval:

$$z_i = -1 + \frac{ih}{s}, \quad i = 0, \dots, Ns, \quad M = Ns + 1. \quad (\text{A.4})$$

We assume $M \geq n$ on the sampled-recovery branch; for fixed s this holds for all sufficiently large admissible widths. The bias is treated as one additional readout coordinate. Thus $n = W + 1$, with feature functions

$$\phi_b(x) = 1, \quad \phi_j(x) = \tanh(\gamma(x - x_j)).$$

The precise reference scaling and truncated-SVD solve are defined in Section A.7. The final computed network is denoted $\hat{q}_W$.

A.1.3 MAIN THEOREM

The next theorem is the formal statement of 3.2 statement. Its explicit numerical quantities are defined in Sections A.7–A.8.

Theorem A.1 (QUILL approximation and finite-precision recovery). *Let $f \in \mathcal{A}_{\delta, B}(I_\Delta)$, fix $0 < \lambda \leq 1$, and let the geometry equation A.1 be admissible. Use the reference-scaled truncated-SVD construction of Section A.7, with a fixed observation density satisfying the sampling conditions in Section A.8. Then the returned readout satisfies*

$$\|\hat{c}_W\|_1 \leq C_{\text{out}}(B, \lambda, \delta, \Delta) + \epsilon_{\text{coeff}, W}, \quad (\text{A.5})$$

and the computed network satisfies

$$\|f - \hat{q}_W\|_{2,\Omega} \leq C_f e^{-c_{\text{app}} W} + \frac{C_f}{W} e^{-\pi^2/\lambda} + \epsilon_{\text{rec},W}, \quad (\text{A.6})$$

where

$$c_{\text{app}} = \frac{1}{8} \min\{\pi\delta, \lambda a_H^2\}. \quad (\text{A.7})$$

Here $\|g\|_{2,\Omega}^2 = \frac{1}{2} \int_{-1}^1 |g(x)|^2 dx$. The constants in the first two terms are independent of width for fixed target class and fixed λ . The quantities $\epsilon_{\text{coeff},W}$ and $\epsilon_{\text{rec},W}$ retain the coefficient-formation, sampling, and floating-point contributions explicitly. The recovery estimate is expressed through the truncation threshold and the bounded comparison vector.

Roadmap. The proof is organized around four subproblems.

- R1. Bounded analytic witness.** Construct q^* in the prescribed tanh dictionary with geometric-plus-replica error and a width-independent readout bound.
- R2. Stable sampled recovery.** Convert the coordinate envelopes of q^* into a bounded Euclidean comparison vector and apply truncated SVD with a cutoff that balances discarded signal against numerical amplification.
- R3. Continuous RMS transfer.** Use local analyticity of the residual to turn sampled RMS control into continuous RMS control at fixed observation density.
- R4. Finite-precision assembly.** Add standard formation and evaluation bounds and isolate the nongeometric allowance.

Each major stage below ends by identifying which roadmap item has been completed.

A.2 LOCAL ANALYTIC TANH REPRESENTATION

Set

$$d = \frac{\pi}{2\gamma}, \quad r = \left(R + \frac{1}{2}\right)h, \quad J = [a, b] = [-1 - r, 1 + r], \quad x_0 = -1. \quad (\text{A.8})$$

The W centers are exactly the midpoints of the width- h cells partitioning J , and $Wh = 2 + 2r$. Define the normalized derivative kernel and the local coefficient density

$$K_\gamma(t) = \frac{\gamma}{2} \text{sech}^2(\gamma t), \quad a_\gamma(z) = \frac{f(z + id) - f(z - id)}{2id} = \frac{1}{2d} \int_{-d}^d f'(z + it) dt. \quad (\text{A.9})$$

The last identity follows by integration along the vertical segment. This density is the local analogue of the derivative weights used in classical integrated-kernel constructions such as Costarelli & Spigler (2015); the point here is that every complex target evaluation remains inside a fixed neighborhood of the finite interval.

[Local density bound] If $\sigma + d < \delta$, then a_γ is holomorphic on a neighborhood of $J + i[-\sigma, \sigma]$, is real on J , and

$$\sup_{t \in J, |y| \leq \sigma} |a_\gamma(t + iy)| \leq M_\sigma := \frac{B}{\delta - \sigma - d}. \quad (\text{A.10})$$

In particular, if $d \leq \delta/2$, then $\|a_\gamma\|_{L^\infty(J)} \leq 2B/\delta$.

Proof. For $t \in J$, $|y| \leq \sigma$, and $|v| \leq d$, the point $t + iy + iv$ is at distance at most $\sigma + d$ from I_Δ . Cauchy's derivative estimate on disks contained in $U_\delta(I_\Delta)$ gives

$$|f'(t + iy + iv)| \leq \frac{B}{\delta - \sigma - d}.$$

Average this bound in equation A.9. Holomorphy follows from the shifted-value formula. Since the target is real on the real interval and its domain is symmetric, Schwarz reflection gives $f(\bar{z}) = \overline{f(z)}$, hence a_γ is real on J . $\square$

For $x \in \Omega$, define

$$D_x(z) = \frac{1}{2} \left[\tanh(\gamma(x-z)) - \tanh(\gamma(x_0-z)) \right], \quad F_x(z) = a_\gamma(z) D_x(z), \quad (\text{A.11})$$

and

$$(\mathcal{C}_h f)(x) = f(x_0) + \int_a^b F_x(t) dt. \quad (\text{A.12})$$

Also set

$$H_x(z) = \coth(\gamma(x-z)) - \coth(\gamma(x_0-z)) \quad (\text{A.13})$$

and

$$(\mathcal{B}_h f)(x) = \frac{1}{4d} \int_{-d}^d \left[f(a+iy) H_x(a+iy) - f(b+iy) H_x(b+iy) \right] dy. \quad (\text{A.14})$$

[Finite local representation] For every $x \in \Omega$,

$$f(x) = \mathcal{C}_h f(x) + \mathcal{B}_h f(x). \quad (\text{A.15})$$

Moreover,

$$\|\mathcal{B}_h f\|_{L^\infty(\Omega)} \leq B [\coth(\gamma r) - \coth(\gamma(r+2))] \leq \frac{2B}{e^{2\gamma r} - 1}. \quad (\text{A.16})$$

Proof. At $x = x_0$ all anchored differences vanish. For $x > x_0$, the choice $\gamma d = \pi/2$ gives

$$H_x(t \pm id) = 2D_x(t).$$

Hence

$$\begin{aligned} \int_a^b F_x(t) dt &= \frac{1}{4id} \left[\int_a^b f(t+id) H_x(t+id) dt \right. \\ &\quad \left. - \int_a^b f(t-id) H_x(t-id) dt \right]. \end{aligned}$$

Apply the residue theorem to $f(z)H_x(z)$ on the rectangle $J + i[-d, d]$. The only interior poles are x and x_0 , with residues $-f(x)/\gamma$ and $f(x_0)/\gamma$, respectively. Since $4d\gamma = 2\pi$, the residue contribution is $-4id[f(x) - f(x_0)]$. Separating horizontal and vertical sides and dividing by $-4id$ gives equation A.15.

For the bound, write $\ell = x - x_0 \in [0, 2]$. The absolutely convergent expansion

$$\coth w = 1 + 2 \sum_{k \geq 1} e^{-2kw}, \quad \Re w > 0,$$

together with oddness on the opposite half-plane gives

$$\begin{aligned} |H_x(a+iy)| &\leq \coth(\gamma r) - \coth(\gamma(r+\ell)), \\ |H_x(b+iy)| &\leq \coth(\gamma(r+2-\ell)) - \coth(\gamma(r+2)). \end{aligned}$$

Each is at most $D_r := \coth(\gamma r) - \coth(\gamma(r+2))$. Substitution in equation A.14 gives $\|\mathcal{B}_h f\|_\infty \leq BD_r$, and $D_r \leq \coth(\gamma r) - 1 = 2/(e^{2\gamma r} - 1)$. $\square$

Remark. The boundary functional in equation A.14 is determined by the two local vertical segments through a and b . We approximate this functional below, which upgrades the halo contribution from equation A.16 to the geometric boundary estimate used in the theorem.

A.3 MIDPOINT DISCRETIZATION AND THE SIGNED FINITE-INTERVAL ERROR

Javed and Trefethen Javed & Trefethen (2014) develop a characteristic-function contour framework for finite-interval trapezoidal and midpoint rules. Applying that framework to the present meromorphic integrand requires retaining the tanh residues. The following short calculation records the resulting signs and endpoint terms.

Let $b = a + Mh$, $t_j = a + (j + \frac{1}{2})h$, $0 \leq j < M$, and

$$Q_h F = h \sum_{j=0}^{M-1} F(t_j), \quad IF = \int_a^b F(t) dt.$$

Suppose F is meromorphic on a neighborhood of $[a, b] + i[-\sigma, \sigma]$, with only simple interior poles and no poles on the boundary or the real segment. Let $\mathcal{P}_+$ and $\mathcal{P}_-$ denote the upper and lower pole sets, and define

$$\zeta_0 = a + \frac{h}{2}, \quad \mathcal{K}_h(z) = \frac{1}{2i} \cot\left(\frac{\pi(z - \zeta_0)}{h}\right), \quad R_h^\pm(z) = \mathcal{K}_h(z) \pm \frac{1}{2}. \quad (\text{A.17})$$

Set

$$\mathcal{H}_h[F] = \int_a^b \left[R_h^-(t - i\sigma) F(t - i\sigma) - R_h^+(t + i\sigma) F(t + i\sigma) \right] dt, \quad (\text{A.18})$$

$$\mathcal{V}_h[F] = i \int_0^\sigma \frac{F(b + iy) - F(a + iy) - F(b - iy) + F(a - iy)}{1 + e^{2\pi y/h}} dy, \quad (\text{A.19})$$

$$\mathcal{P}_h[F] = 2\pi i \left[\sum_{p \in \mathcal{P}_+} R_h^+(p) \text{Res}(F; p) + \sum_{p \in \mathcal{P}_-} R_h^-(p) \text{Res}(F; p) \right]. \quad (\text{A.20})$$

[Signed meromorphic midpoint identity] With the orientations above,

$$Q_h F - IF = \mathcal{H}_h[F] + \mathcal{V}_h[F] - \mathcal{P}_h[F]. \quad (\text{A.21})$$

Moreover,

$$|R_h^+(z)| \leq \frac{1}{e^{2\pi \Im z/h} - 1} \quad (\Im z > 0), \quad |R_h^-(z)| \leq \frac{1}{e^{2\pi |\Im z|/h} - 1} \quad (\Im z < 0). \quad (\text{A.22})$$

Proof. Let Γ_+ traverse $b \rightarrow b + i\sigma \rightarrow a + i\sigma \rightarrow a$, and let Γ_- traverse $a \rightarrow a - i\sigma \rightarrow b - i\sigma \rightarrow b$. The residue theorem gives

$$\int_{\Gamma_+} F = 2\pi i \sum_{p \in \mathcal{P}_+} \text{Res}(F; p) - IF, \quad \int_{\Gamma_-} F = 2\pi i \sum_{p \in \mathcal{P}_-} \text{Res}(F; p) + IF.$$

Each grid pole of $\mathcal{K}_h$ has residue $h/(2\pi i)$, so on the full rectangle

$$\int_{\Gamma_+ \cup \Gamma_-} \mathcal{K}_h F = Q_h F + 2\pi i \sum_{p \in \mathcal{P}_+ \cup \mathcal{P}_-} \mathcal{K}_h(p) \text{Res}(F; p).$$

Substitute $\mathcal{K}_h = R_h^+ - 1/2$ on Γ_+ and $\mathcal{K}_h = R_h^- + 1/2$ on Γ_- . The horizontal pieces give equation A.18. At the half-cell endpoints,

$$R_h^+(a + iy) = R_h^+(b + iy) = \frac{1}{1 + e^{2\pi y/h}}, \quad R_h^-(a - iy) = R_h^-(b - iy) = -\frac{1}{1 + e^{2\pi y/h}},$$

which gives equation A.19. This proves equation A.21. Finally, if $w = e^{2\pi i(z - \zeta_0)/h}$, then $R_h^+(z) = w/(w-1)$ and $R_h^-(z) = 1/(w-1)$, from which equation A.22 follows by the triangle inequality. $\square$

Return to the QUILL geometry and choose

$$k = \left\lceil \frac{\delta}{8d} \right\rceil, \quad \sigma = 2kd. \quad (\text{A.23})$$

Then, under $d \leq \delta/8$,

$$\frac{\delta}{4} \leq \sigma \leq \frac{\delta}{2}, \quad \sigma + d \leq \frac{5\delta}{8}, \quad M_\sigma \leq \frac{4B}{\delta}. \quad (\text{A.24})$$

Also $\gamma\sigma = k\pi$, so $D_x(t \pm i\sigma) = D_x(t)$, while the contour avoids the odd tanh pole layers.

[Tanh quadrature contributions] Uniformly for $x \in \Omega$,

$$|\mathcal{H}_h[F_x]| \leq \frac{4M_\sigma}{e^{2\pi\sigma/h} - 1}, \quad (\text{A.25})$$

$$|\mathcal{P}_h[F_x]| \leq \frac{4\pi M_\sigma}{\gamma} \sum_{\ell=0}^{k-1} \frac{1}{e^{(2\ell+1)\pi^2/\lambda} - 1}, \quad (\text{A.26})$$

$$|\mathcal{V}_h[F_x]| \leq \frac{4M_\sigma h \log 2}{\pi(e^{2\gamma r} - 1)}. \quad (\text{A.27})$$

Proof. For real t ,

$$D_x(t) = \int_{x_0}^x K_\gamma(v - t) dv \geq 0, \quad \int_{\mathbb{R}} D_x(t) dt = x - x_0 \leq 2,$$

because K_γ has unit mass. Therefore $\int_J |F_x(t \pm i\sigma)| dt \leq 2M_\sigma$, and equation A.25 follows from equation A.22.

The possible poles are

$$x \pm i(2\ell + 1)d, \quad x_0 \pm i(2\ell + 1)d, \quad 0 \leq \ell < k.$$

Their residues in D_x are $\mp 1/(2\gamma)$, so each residue of F_x has modulus at most $M_\sigma/(2\gamma)$. Since

$$\frac{2\pi(2\ell + 1)d}{h} = \frac{(2\ell + 1)\pi^2}{\lambda},$$

equation A.20 and equation A.22 give equation A.26.

At either vertical endpoint, subtract the common saturated value of the two tanhs and use

$$\tanh w - 1 = -\frac{2}{e^{2w} + 1}, \quad \Re w > 0,$$

together with oddness on the left half-plane. This yields

$$|D_x(a \pm iy)|, |D_x(b \pm iy)| \leq \frac{2}{e^{2\gamma r} - 1}.$$

Substitute in equation A.19 and use $\int_0^\infty (1 + e^{2\pi y/h})^{-1} dy = h \log 2/(2\pi)$ to obtain equation A.27. $\square$

For later convenience define

$$\mathcal{S}(A) = \sum_{\ell \geq 0} \frac{1}{e^{(2\ell+1)A} - 1}, \quad \mathcal{U}(A) = \frac{e^{-A}}{(1 - e^{-A})(1 - e^{-2A})}. \quad (\text{A.28})$$

Then $\mathcal{S}(A) \leq \mathcal{U}(A)$, and $\mathcal{U}(A) \leq 2e^{-A}$ for $A \geq \pi^2$.

The midpoint rule applied to F_x is exactly the prescribed tanh network

$$q_h(x) = f(x_0) + h \sum_{j=-R}^{N+R} a_\gamma(x_j) D_x(x_j). \quad (\text{A.29})$$

Its baseline tanh weights are $w_j^{\text{base}} = ha_\gamma(x_j)/2$, with the constant terms absorbed into the anchored bias.

[Baseline signed error] The network equation A.29 satisfies

$$f - q_h = \mathcal{B}_h f - \mathcal{V}_h[F_x] + \mathcal{P}_h[F_x] - \mathcal{H}_h[F_x]. \quad (\text{A.30})$$

Moreover,

$$|w_j^{\text{base}}| \leq \frac{Bh}{\delta}, \quad \sum_j |w_j^{\text{base}}| \leq \frac{2B(1 + \Delta)}{\delta}, \quad |b^{\text{base}}| \leq B + \frac{2B(1 + \Delta)}{\delta}. \quad (\text{A.31})$$

Proof. Subtract Lemma A.3 from the exact identity in Lemma A.2. The coefficient bounds use Lemma A.2, $Wh = 2 + 2r \leq 2 + 2\Delta$, and the anchored bias. $\square$

The first two terms in equation A.30 are both endpoint effects. Define their combined contribution

$$\mathcal{B}_h^{\text{eff}} f(x) := \mathcal{B}_h f(x) - \mathcal{V}_h[F_x]. \quad (\text{A.32})$$

The effective boundary functional combines the finite-representation boundary contribution with the midpoint endpoint contribution; the correction below treats them together.

Roadmap checkpoint. The local representation and midpoint analysis reduce **R1** to one concrete task: approximate the effective boundary functional $\mathcal{B}_h^{\text{eff}} f$ using the existing outer halo while retaining coefficient control. The remaining horizontal-contour and tanh-replica terms already have explicit bounds.

A.4 BOUNDARY CORRECTION USING THE EXISTING HALO

Set

$$z_-(x) = e^{-2\gamma(x-a)}, \quad z_+(x) = e^{-2\gamma(b-x)}, \quad \theta = e^{-\gamma r}, \quad z_{\pm,0} = z_{\pm}(x_0). \quad (\text{A.33})$$

For $x \in \Omega$, all of these arguments lie in $(0, \theta^2]$. For $\ell \geq 1$, define

$$\mu_\ell^- = \frac{1}{2d} \int_{-d}^d f(a + iy) e^{2i\ell\gamma y} dy, \quad \mu_\ell^+ = \frac{1}{2d} \int_{-d}^d f(b + iy) e^{-2i\ell\gamma y} dy, \quad (\text{A.34})$$

$$\nu_\ell^- = i(-1)^\ell \int_0^\sigma \frac{a_\gamma(a - iy) e^{-2i\ell\gamma y} - a_\gamma(a + iy) e^{2i\ell\gamma y}}{1 + e^{2\pi y/h}} dy, \quad (\text{A.35})$$

$$\nu_\ell^+ = i(-1)^\ell \int_0^\sigma \frac{a_\gamma(b - iy) e^{2i\ell\gamma y} - a_\gamma(b + iy) e^{-2i\ell\gamma y}}{1 + e^{2\pi y/h}} dy. \quad (\text{A.36})$$

Put

$$g_\ell^\pm = \mu_\ell^\pm - \nu_\ell^\pm, \quad G^\pm(z) = \sum_{\ell \geq 1} g_\ell^\pm z^\ell, \quad D_\lambda = \frac{\pi}{2\lambda} + \frac{4 \log 2}{\pi}. \quad (\text{A.37})$$

[Boundary series and the extra factor h] The coefficients $g_\ell^\pm$ are real and satisfy

$$|g_\ell^\pm| \leq M_h := \frac{Bh}{\delta} D_\lambda. \quad (\text{A.38})$$

The functions $G^\pm$ are holomorphic on $|z| < 1$, and

$$\begin{aligned} \mathcal{B}_h^{\text{eff}} f(x) &= G^-(z_-(x)) - G^-(z_{-,0}) \\ &\quad + G^+(z_+(x)) - G^+(z_{+,0}). \end{aligned} \quad (\text{A.39})$$

Proof. For $\Re w > 0$,

$$\coth w = 1 + 2 \sum_{\ell \geq 1} e^{-2\ell w}, \quad \tanh w = 1 + 2 \sum_{\ell \geq 1} (-1)^\ell e^{-2\ell w},$$

and use oddness at the right boundary. With $\Delta z_\pm^\ell = z_\pm(x)^\ell - z_{\pm,0}^\ell$, this gives uniformly convergent endpoint expansions

$$\begin{aligned} H_x(a + iy) &= 2 \sum_{\ell \geq 1} \Delta z_-^\ell e^{2i\ell\gamma y}, & H_x(b + iy) &= -2 \sum_{\ell \geq 1} \Delta z_+^\ell e^{-2i\ell\gamma y}, \\ D_x(a + iy) &= \sum_{\ell \geq 1} (-1)^\ell \Delta z_-^\ell e^{2i\ell\gamma y}, & D_x(b + iy) &= - \sum_{\ell \geq 1} (-1)^\ell \Delta z_+^\ell e^{-2i\ell\gamma y}. \end{aligned}$$

Termwise integration in equation A.14 and equation A.19 yields equation A.39.

The sharper coefficient bound is the key step. Since $\gamma d = \pi/2$,

$$\int_{-d}^d e^{2i\ell\gamma y} dy = 0, \quad \ell \geq 1. \quad (\text{A.40})$$

Subtract $f(a)$ in μ_ℓ^- and $f(b)$ in μ_ℓ^+ . Cauchy's estimate gives $|f'(a + it)|, |f'(b + it)| \leq 2B/\delta$ for $|t| \leq d$, and therefore

$$|\mu_\ell^\pm| \leq \frac{1}{2d} \frac{2B}{\delta} \int_{-d}^d |y| dy = \frac{Bd}{\delta} = \frac{\pi Bh}{2\lambda\delta}.$$

Likewise Lemma A.2, equation A.24, and the Fermi-type endpoint weight give

$$|\nu_\ell^\pm| \leq 2M_\sigma \int_0^\sigma \frac{dy}{1 + e^{2\pi y/h}} \leq \frac{4Bh \log 2}{\pi\delta}.$$

This proves equation A.38. Conjugate symmetry makes the coefficients real. Bounded coefficients imply holomorphy of $G^\pm$ on the unit disk. $\square$

Why the $O(h)$ cancellation matters. The cancellation equation A.40 supplies the additional factor h . This factor is what allows a sublinear halo to yield a geometric-in-width boundary estimate while keeping the readout controlled.

A.4.1 A PRESCRIBED-DENOMINATOR RATIONAL CORRECTION

The next construction is a standard Padé-type idea: prescribe the denominator poles and choose the numerator so that the Taylor series matches through the desired order. See Brezinski and Redivo-Zaglia Brezinski & Redivo-Zaglia (2015) for general Padé-type and partial-fraction representations. We record the short specialization needed here because its error estimate makes the halo rate transparent.

[Prescribed-denominator approximation] Let $0 < \theta < 1$ and $G(z) = \sum_{\ell \geq 1} g_\ell z^\ell$ with $|g_\ell| \leq M$. Choose distinct $0 < t_i \leq \theta^{-1}$, $1 \leq i \leq m$, and define

$$D_m(z) = \prod_{i=1}^m (1 + t_i z), \quad P_m = [D_m G]_{\leq m}, \quad S_m = \frac{P_m}{D_m}. \quad (\text{A.41})$$

Then

$$\sup_{0 \leq z \leq \theta^2} |G(z) - S_m(z)| \leq \frac{M\theta}{(1-\theta)^2} (2\theta)^m. \quad (\text{A.42})$$

Moreover $S_m = c_0 + \sum_{i=1}^m c_i/(1 + t_i z)$, with real coefficients when the data are real.

Proof. On $|z| = \theta$ we use the conservative bounds

$$|G(z)| \leq \frac{M}{1-\theta}, \quad |D_m(z)| \leq \prod_i (1 + t_i \theta) \leq 2^m.$$

Cauchy's coefficient estimate therefore gives

$$|[z^k](D_m G)| \leq \frac{M 2^m \theta^{-k}}{1-\theta}.$$

For $0 \leq z \leq \theta^2$, summing the tail beyond degree m gives

$$\frac{M 2^m}{1-\theta} \sum_{k \geq m+1} \theta^k = \frac{M\theta}{(1-\theta)^2} (2\theta)^m.$$

Since $D_m(z) \geq 1$ on the positive real axis, division yields equation A.42. The partial-fraction form follows from distinct denominator roots. $\square$

Choose $1 \leq m \leq \lfloor (R+1)/2 \rfloor$ and the m outermost centers on each side:

$$\xi_i^- = a + \left(i - \frac{1}{2}\right)h, \quad \xi_i^+ = b - \left(i - \frac{1}{2}\right)h, \quad t_i = e^{2\lambda(i-1/2)}. \quad (\text{A.43})$$

These centers lie in the outer halves of the halos, and $t_i \leq e^{\gamma r} = \theta^{-1}$. The key identities are

$$\frac{1}{1 + t_i z_-(x)} = \frac{1 + \tanh(\gamma(x - \xi_i^-))}{2}, \quad \frac{1}{1 + t_i z_+(x)} = \frac{1 - \tanh(\gamma(x - \xi_i^+))}{2}. \quad (\text{A.44})$$

Thus the rational correction is exactly representable in the existing tanh dictionary.

Apply Lemma A.4.1 to G^- and G^+ , and define

$$q_{\text{bd}}(x) = S_m^-(z_-(x)) - S_m^-(z_{-,0}) + S_m^+(z_+(x)) - S_m^+(z_{+,0}). \quad (\text{A.45})$$

[Boundary correction in the same dictionary] The corrected network

$$q^* = q_h + q_{\text{bd}} \quad (\text{A.46})$$

uses the same W hidden-unit slots as the baseline network and satisfies

$$\|\mathcal{B}_h^{\text{eff}} f - q_{\text{bd}}\|_{L^\infty(\Omega)} \leq \frac{4M_h\theta}{(1-\theta)^2} (2\theta)^m. \quad (\text{A.47})$$

If $m = \lfloor (R+1)/2 \rfloor$ and $\lambda R \geq 2 \log 2$, then

$$\|\mathcal{B}_h^{\text{eff}} f - q_{\text{bd}}\|_\infty \leq 16M_h e^{-\lambda R^2/4}. \quad (\text{A.48})$$

Proof. Apply equation A.42 at the four arguments in the two anchored differences in equation A.39. Identity equation A.44 realizes every partial fraction using a selected halo tanh feature; the constants cancel in each anchored difference. This proves equation A.47.

For the simplified bound, $m \geq R/2$, $\theta = e^{-\lambda(R+1/2)} \leq e^{-\lambda R} \leq 1/4$, and

$$(2\theta)^m = e^{-m(\lambda(R+1/2) - \log 2)} \leq e^{-(R/2)(\lambda R/2)} = e^{-\lambda R^2/4}.$$

Also $4\theta/(1-\theta)^2 \leq 16$, giving equation A.48. $\square$

A.5 WIDTH-INDEPENDENT READOUT CONTROL

The partial-fraction coefficient map is an inverse-Vandermonde map; classical explicit formulas and norm estimates are given by Gautschi Gautschi (1962). We specialize that algebra to the geometric nodes equation A.43, where the resulting row sums are uniformly summable in width.

Let

$$\zeta = e^{-2\lambda}, \quad r_i = t_i^{-1} = \zeta^{i-1/2}, \quad p_k = \prod_{j=1}^k (1 - \zeta^j), \quad p_0 = 1. \quad (\text{A.49})$$

For a finite list, let e_k denote the elementary symmetric polynomial of degree k .

[Scaled partial-fraction coefficients] For either boundary series $G(z) = \sum_{\ell \geq 1} g_\ell z^\ell$, the nonconstant coefficients in the partial-fraction expansion of S_m satisfy

$$c_i = (-1)^i \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1}p_{m-i}} \sum_{\ell=1}^m e_{\ell-1}(r_1, \dots, \hat{r}_i, \dots, r_m) g_\ell. \quad (\text{A.50})$$

Proof. The classical partial-fraction formula gives

$$c_i = \frac{P_m(-1/t_i)}{\prod_{j \neq i} (1 - t_j/t_i)}.$$

Equivalently, this is a row of the inverse Vandermonde map associated with the prescribed poles; see Gautschi (1962). Factor $D_m = (1 + t_i z) D_{m,i}$ and expand $D_{m,i}$ in elementary symmetric polynomials. Evaluating the truncated product at $z = -r_i$ telescopes the contribution of $(1 + t_i z)$, leaving

$$P_m(-r_i) = (-r_i)^m \sum_{\ell=1}^m g_\ell d_{m-\ell}^{(i)}, \quad d_{m-\ell}^{(i)} = \left(\prod_{j \neq i} t_j \right) e_{\ell-1}(r_{\neq i}).$$

For the geometric nodes,

$$\prod_{j \neq i} (1 - t_j/t_i) = (-1)^{m-i} \zeta^{-(m-i)(m-i+1)/2} p_{i-1} p_{m-i}.$$

Collecting the sign and powers of ζ gives equation A.50. $\square$

Define

$$p_\infty = \prod_{j \geq 1} (1 - \zeta^j), \quad L_+ = \prod_{j \geq 1} (1 + \zeta^{j-1/2}), \quad (\text{A.51})$$

and

$$C_{\text{bd}}(\lambda) = \sum_{i \geq 1} \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1} p_\infty} \prod_{\substack{j \geq 1 \\ j \neq i}} (1 + \zeta^{j-1/2}). \quad (\text{A.52})$$

[Width-independent boundary amplification] For every fixed $\lambda > 0$, $0 < C_{\text{bd}}(\lambda) < \infty$, and the partial-fraction coefficients satisfy

$$\sum_{i=1}^m |c_i| \leq C_{\text{bd}}(\lambda) \|g\|_{\ell_\infty}, \quad (\text{A.53})$$

where $\|g\|_{\ell_\infty} = \sup_{\ell \geq 1} |g_\ell|$. In particular, the bound is independent of m . A convenient elementary envelope is

$$C_{\text{bd}}(\lambda) \leq \frac{\sqrt{\pi} e^\lambda}{2\sqrt{\lambda}} \exp\left(\frac{5\pi^2}{24\lambda}\right). \quad (\text{A.54})$$

Proof. All elementary symmetric coefficients in equation A.50 are nonnegative, and

$$\sum_{\ell=1}^m e_{\ell-1}(r_{\neq i}) = \prod_{j \neq i} (1 + r_j).$$

Also $p_{m-i} \geq p_\infty > 0$. Therefore the absolute row sum in equation A.50 is bounded by the i -th summand of equation A.52. Since $\sum_j \zeta^j < \infty$, taking logarithms shows $p_\infty > 0$ and $L_+ < \infty$. The remaining factor decays quadratically in i in the exponent, so the sum converges.

For equation A.54,

$$\log p_\infty^{-1} = \sum_{k \geq 1} \frac{1}{k(e^{2\lambda k} - 1)} \leq \frac{1}{2\lambda} \sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{12\lambda}.$$

The function $t \mapsto \log(1 + e^{-t})$ is convex, so midpoint comparison with the integral gives

$$\log L_+ \leq \frac{1}{2\lambda} \int_0^\infty \log(1 + e^{-t}) dt = \frac{\pi^2}{24\lambda}.$$

Finally,

$$\sum_{i \geq 1} \zeta^{[i(i+1)-1]/2} = e^\lambda \sum_{i \geq 1} e^{-\lambda i(i+1)} \leq e^\lambda \int_0^\infty e^{-\lambda t^2} dt = \frac{e^\lambda \sqrt{\pi}}{2\sqrt{\lambda}}.$$

Multiplying the three bounds yields equation A.54. $\square$

[Bounded corrected readout] For the corrected reference network q^* ,

$$\sum_j |w_j^*| \leq \frac{2B(1+\Delta)}{\delta} + C_{\text{bd}}(\lambda) \frac{Bh}{\delta} D_\lambda, \quad (\text{A.55})$$

and

$$\|c^*\|_1 \leq B \left[1 + \frac{4(1+\Delta)}{\delta} + \frac{2C_{\text{bd}}(\lambda)D_\lambda h}{\delta} \right]. \quad (\text{A.56})$$

In particular, for admissible widths one may use the common width-independent cap

$$K(\lambda, \delta, \Delta) = 1 + \frac{4(1+\Delta)}{\delta} + \frac{C_{\text{bd}}(\lambda)D_\lambda}{4\delta}, \quad \|c^*\|_1 \leq KB. \quad (\text{A.57})$$

Proof. Each boundary contributes one half of its partial-fraction absolute sum to the tanh readout by equation A.44. Combine Lemma A.4, Proposition A.5, and the baseline bound equation A.31. The anchored bias is at most $B + \sum_j |w_j^*|$. Finally, admissibility gives $N \geq 16$ and therefore $h \leq 1/8$, yielding equation A.57. $\square$

A.6 THE BOUNDED ANALYTIC WITNESS

Combining the signed decomposition with the correction gives

$$f - q^* = (\mathcal{B}_h^{\text{eff}} f - q_{\text{bd}}) + \mathcal{P}_h[F_x] - \mathcal{H}_h[F_x]. \quad (\text{A.58})$$

The uncompressed bound is

$$\begin{aligned} \|f - q^*\|_\infty &\leq \frac{16B}{\delta(e^{\pi\delta/(2h)} - 1)} + \frac{16\pi Bh}{\lambda\delta} \mathcal{U}(\pi^2/\lambda) \\ &\quad + \frac{4BhD_\lambda}{\delta} \frac{\theta}{(1-\theta)^2} (2\theta)^m, \quad \theta = e^{-\lambda(R+1/2)}. \end{aligned} \quad (\text{A.59})$$

This follows directly from Lemma A.3, equation A.28, and Proposition A.4.1.

Theorem A.2 (Boundary-corrected analytic witness). *For every $f \in \mathcal{A}_{\delta,B}(I_\Delta)$ and every admissible width, the explicit network q^* satisfies*

$$\|f - q^*\|_{L^\infty(\Omega)} \leq \frac{128\pi B}{\lambda\delta} \left[e^{-c_{\text{app}}W} + \frac{e^{-\pi^2/\lambda}}{W} \right] =: E_{\text{rep}}(W, \lambda), \quad (\text{A.60})$$

where c_{app} is given in equation A.7, and

$$\|c^*\|_1 \leq K(\lambda, \delta, \Delta)B. \quad (\text{A.61})$$

Thus $R = O(\sqrt{N})$, $W = N + O(\sqrt{N})$, and $\gamma = \Theta(W)$ for fixed λ .

Proof. Take $m = \lfloor (R+1)/2 \rfloor$. The first term in equation A.59 is at most

$$\frac{32B}{\delta} e^{-\pi\delta N/4},$$

because $h = 2/N$, $\gamma\delta \geq 4\pi$, and $\lambda \leq 1$ make the exponent safely larger than $\log 2$. Proposition A.4.1 and $R \geq a_H \sqrt{N}$ give

$$\text{boundary term} \leq \frac{32BD_\lambda}{\delta N} e^{-\lambda a_H^2 N/4}.$$

Set

$$\kappa = \frac{1}{4} \min\{\pi\delta, \lambda a_H^2\}.$$

Admissibility implies

$$N \leq W = N + 2R + 1 \leq 2N,$$

and $\lambda(1 + D_\lambda) < 4\pi$. Hence the resolution and corrected-boundary terms are bounded by

$$\frac{128\pi B}{\lambda\delta} e^{-\kappa W/2} = \frac{128\pi B}{\lambda\delta} e^{-c_{\text{app}}W}.$$

For the pole replicas, $\mathcal{U}(A) \leq 2e^{-A}$ at $A = \pi^2/\lambda$ yields

$$\frac{16\pi Bh}{\lambda\delta} \mathcal{U}(\pi^2/\lambda) \leq \frac{64\pi B}{\lambda\delta N} e^{-\pi^2/\lambda} \leq \frac{128\pi B}{\lambda\delta W} e^{-\pi^2/\lambda}.$$

This proves equation A.60. The coefficient bound is Proposition A.5. $\square$

Roadmap checkpoint. R1 is complete. We now have a comparison vector c^* in the prescribed dictionary with geometric-plus-replica approximation error and a total readout bound independent of width. The recovery stage uses this network through its approximation bound and coordinate envelopes.

A.7 REFERENCE-SCALED TRUNCATED-SVD RECOVERY

The analytic construction supplies a bounded comparison network in the prescribed feature space. We now use its coordinate envelopes to scale the sampled least-squares system, then apply a standard truncated-SVD comparison argument.

A.7.1 COORDINATE ENVELOPES AND REFERENCE SCALING

For the numerical comparison, retain coordinatewise versions of the coefficient bounds. Put

$$d_{\text{pole}} = \frac{\pi h}{2\lambda} = d, \quad m = \left\lfloor \frac{R+1}{2} \right\rfloor, \quad \zeta = e^{-2\lambda}. \quad (\text{A.62})$$

Define

$$L_{i,m} = \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1}p_{m-i}} \prod_{\substack{1 \leq j \leq m \\ j \neq i}} (1 + \zeta^{j-1/2}), \quad C_m = \sum_{i=1}^m L_{i,m}. \quad (\text{A.63})$$

For every output-weight slot, take the baseline envelope

$$\alpha_j^{\text{base}} = \frac{h}{2(\delta - d_{\text{pole}})}, \quad (\text{A.64})$$

and add, at corrected slot i on either side,

$$\alpha_i^{\text{bd}} = \frac{h D_\lambda L_{i,m}}{2\delta}. \quad (\text{A.65})$$

Let α_j denote the resulting weight envelopes and set

$$\alpha_b = 1 + \sum_{\text{weight slots } j} \alpha_j. \quad (\text{A.66})$$

The construction above gives

$$|c_j^*| \leq B\alpha_j \quad \text{for every readout coordinate.} \quad (\text{A.67})$$

Writing

$$K_W^{\text{ref}} := \sum_j \alpha_j = 1 + \frac{Wh}{\delta - d_{\text{pole}}} + \frac{2hD_\lambda C_m}{\delta}, \quad (\text{A.68})$$

Proposition A.5 implies $K_W^{\text{ref}} \leq K_{\text{ref}}$ uniformly over admissible widths for fixed $(\lambda, \delta, \Delta)$.

Choose the reference scale

$$D = \text{diag}(d_j), \quad d_j = \sqrt{\alpha_j}, \quad L = \|d\|_2 = (K_W^{\text{ref}})^{1/2}, \quad S = BL. \quad (\text{A.69})$$

Let $A \in \mathbb{R}^{M \times n}$ be the exact sampled feature matrix, including the bias column, and define

$$F = \frac{AD}{\sqrt{M}}, \quad y_X = \frac{(f(z_i))_{i=0}^{M-1}}{\sqrt{M}}, \quad a^* = D^{-1}c^*. \quad (\text{A.70})$$

[Bounded comparison vector in scaled coordinates] The reference vector satisfies

$$\|a^*\|_2 \leq S, \quad \|Fa^* - y_X\|_2 \leq E_{\text{rep}}(W, \lambda), \quad (\text{A.71})$$

and, for every $a \in \mathbb{R}^n$,

$$\|Da\|_1 \leq L \|a\|_2. \quad (\text{A.72})$$

Proof. By equation A.67 and $d_j^2 = \alpha_j$,

$$\|a^*\|_2^2 = \sum_j \frac{|c_j^*|^2}{d_j^2} \leq B^2 \sum_j \alpha_j = B^2 L^2 = S^2.$$

The sampled residual bound follows from Theorem A.2. Cauchy–Schwarz gives $\|Da\|_1 \leq \|d\|_2 \|a\|_2 = L \|a\|_2$. $\square$

A.7.2 FINITE-PRECISION TRUNCATED-SVD INTERFACE

We use standard floating-point results for dot products and singular-value decompositions. Higham (Higham, 2002, Chs. 2–4) gives the operationwise rounding model and pairwise-summation analysis, and the LAPACK Users’ Guide Anderson et al. (1999) gives backward-error bounds for computed singular value decompositions. A closely related approximation setting is the polynomial-frame construction of Adcock and Shadrin Adcock & Shadrin (2022), which combines equispaced samples, oversampled least squares, and SVD regularization to obtain geometric refinement to a regularization-controlled tolerance. Here the approximation space is the prescribed tanh dictionary, and the bounded-reference cutoff estimate used for QUILL is proved directly below.

We package these standard estimates in the form needed below. Let $\tilde{F}$ and $\tilde{y}$ denote the assembled working-precision matrix and labels, and let the computed SVD be truncated at a threshold τ . There are exact comparison objects F_0, y_0 and an exact truncated-SVD coefficient vector a_τ for the nearby problem (F_0, y_0) such that

$$\|F - F_0\|_2 \leq a_W, \quad \|y_X - y_0\|_2 \leq b_W. \quad (\text{A.73})$$

The quantities a_W and b_W collect feature assembly, label formation, factorization, projection, and retained-coordinate division errors. Standard pairwise dot-product bounds have the form

$$|\text{pdot}(v, w) - v^T w| \leq \Gamma_k^{\text{pw}} \sum_{i=1}^k |v_i w_i| + \beta_k^{\text{pw}}, \quad \Gamma_k^{\text{pw}} = O(u \log k), \quad (\text{A.74})$$

with β_k^{pw} accounting for subnormal absolute errors when they are retained in the arithmetic model. At fixed observation density, $M = \Theta(W)$ and $n = \Theta(W)$; applying equation A.74 to the n projected coordinates gives the $O(u\sqrt{W} \log W)$ contribution highlighted in the main text, up to the corresponding matrix/vector scales.

Likewise, coefficient export from the retained scaled coordinates gives a computable allowance $e_{c,W}$ satisfying

$$\|\hat{c} - Da_\tau\|_1 \leq e_{c,W}. \quad (\text{A.75})$$

The stored hidden geometry may be represented exactly, as in the dyadic specialization $\lambda = 1/4$, $N = 2^k$, $s = 2^\ell$, or its validated storage error may be included in a_W . These quantities enter the explicit implementation allowance and remain separate from the analytic approximation constants.

For final network evaluation, let ϵ_ϕ bound the absolute error of one stored tanh feature evaluated with its prescribed affine argument, and use the same pairwise bound equation A.74 for the readout sum. These standard arithmetic quantities enter the final error formula in Section A.8.

A.7.3 ADAPTIVE CUTOFF AND BOUNDED RECOVERY

Set

$$\eta_W = E_{\text{rep}}(W, \lambda) + a_W S + b_W. \quad (\text{A.76})$$

Choose the truncation threshold so that

$$\tau S \geq \eta_W + \epsilon_{\text{reg}}, \quad (\text{A.77})$$

where ϵ_{reg} accounts for forming the cutoff in working precision, and choose

$$\rho_W \geq \tau S - \eta_W. \quad (\text{A.78})$$

[Reference-scaled truncated-SVD recovery] Under equation A.71, equation A.73, and equation A.77, the exact truncated-SVD solution of the nearby problem satisfies

$$\|a_\tau\|_2 \leq 2S, \quad (\text{A.79})$$

and

$$\|Fa_\tau - y_X\|_2 \leq 2E_{\text{rep}}(W, \lambda) + 4a_W S + 3b_W + \rho_W. \quad (\text{A.80})$$

Proof. By equation A.71 and equation A.73,

$$\|F_0 a^\star - y_0\|_2 \leq \eta_W.$$

Write $y_0 = F_0 a^* + e$, $\|e\|_2 \leq \eta_W$, and let P_τ project onto the retained right singular subspace of F_0 . Exact truncated SVD gives

$$a_\tau = P_\tau a^* + (F_0)_\tau^\dagger e. \quad (\text{A.81})$$

Hence

$$\|a_\tau\|_2 \leq S + \frac{\eta_W}{\tau} \leq 2S.$$

The discarded component of a^* contributes at most τS , so

$$\|F_0 a_\tau - y_0\|_2 \leq \eta_W + \tau S.$$

Returning to the original sampled problem gives

$$\begin{aligned} \|F a_\tau - y_X\|_2 &\leq \eta_W + \tau S + a_W \|a_\tau\|_2 + b_W \\ &\leq E_{\text{rep}}(W, \lambda) + \tau S + 3a_W S + 2b_W \\ &\leq 2E_{\text{rep}}(W, \lambda) + 4a_W S + 3b_W + \rho_W. \end{aligned}$$

□

The two inequalities in Proposition A.7.3 expose the cutoff mechanism directly: retained singular directions amplify the perturbation by at most $1/\tau$, while the discarded part of the bounded comparison vector contributes at most τS . The reference scaling makes both terms measurable in the same Euclidean coordinate norm.

Combining equation A.72, equation A.79, and equation A.75, define

$$T = 2LS + e_{c,W}, \quad R_X = 2E_{\text{rep}}(W, \lambda) + 4a_W S + 3b_W + \rho_W + e_{c,W}. \quad (\text{A.82})$$

Since $|\phi_j(x)| \leq 1$ on the real axis,

$$\|\hat{c}\|_1 \leq T, \quad \|f - q_{\hat{c}}\|_{2,X} \leq R_X, \quad (\text{A.83})$$

where $\|g\|_{2,X}^2 = M^{-1} \sum_i |g(z_i)|^2$. Since $L^2 = K_W^{\text{ref}} \leq K_{\text{ref}}$, the reference part of the readout bound is at most $2BK_{\text{ref}}$ uniformly in width.

Roadmap checkpoint. R2 is complete. The sampled truncated-SVD output has a controlled sampled RMS residual and a controlled native readout. The construction-specific inputs are the coordinate envelopes and the cutoff balance; the floating-point factorization and dot-product estimates are standard numerical-analysis ingredients.

A.8 FROM SAMPLED RMS TO CONTINUOUS RMS

We next transfer equation A.83 to the continuous norm used in Theorem A.1. This is where analyticity of the target and the bounded recovered readout are used together.

Let $q = q_{\hat{c}}$ denote the exact real function defined by the stored coefficients and the prescribed hidden geometry, and let

$$g = f - q.$$

Fix a Bernstein-ellipse parameter $\varrho > 1$ and an integer polynomial degree $p \geq 1$, and assume

$$\lambda(\varrho - \varrho^{-1}) \leq \pi, \quad \frac{h}{4}(\varrho + \varrho^{-1}) < \delta, \quad s \geq 2(p+1)^2. \quad (\text{A.84})$$

On each core cell, the closed Bernstein ellipse of parameter ϱ remains in the target tube and in the strip where all tanh features are bounded by one. Indeed,

$$|\tanh(u + iv)|^2 = \frac{\sinh^2 u + \sin^2 v}{\sinh^2 u + \cos^2 v} \leq 1 \quad (|v| \leq \pi/4), \quad (\text{A.85})$$

and the vertical semiaxis in preactivation coordinates is at most $\lambda(\varrho - \varrho^{-1})/4 \leq \pi/4$. Hence

$$|g(z)| \leq B + T \quad (\text{A.86})$$

on every such ellipse.

For a function bounded by M on a Bernstein ellipse of parameter ϱ , the standard Chebyshev/Bernstein estimate gives a degree- p polynomial P satisfying

$$\|g - P\|_{L^\infty(I)} \leq \frac{2M}{\varrho - 1} \varrho^{-p}. \quad (\text{A.87})$$

See Trefethen (Trefethen, 2019, Ch. 8). Applying equation A.87 with $M = B + T$ gives, on each cell I ,

$$e_p := \|g - P\|_{L^\infty(I)} \leq \frac{2(B + T)}{\varrho - 1} \varrho^{-p}. \quad (\text{A.88})$$

For a degree- p polynomial, the classical Legendre-basis inverse estimate gives the bound below. With the normalized cell norm

$$\|P\|_{2,I}^2 = \frac{1}{h} \int_I |P(x)|^2 dx,$$

one convenient bound is

$$\|P'\|_{2,I} \leq \frac{(p+1)\sqrt{p(p+2)}}{h} \|P\|_{2,I}. \quad (\text{A.89})$$

Let $\|P\|_{\text{trap},I}$ be the composite trapezoidal norm on the $s+1$ cell sample points. Applying the elementary trapezoidal remainder estimate to $w = |P|^2$, followed by Cauchy–Schwarz and equation A.89, gives

$$\left| \|P\|_{\text{trap},I}^2 - \|P\|_{2,I}^2 \right| \leq \frac{(p+1)\sqrt{p(p+2)}}{s} \|P\|_{2,I}^2 \leq \frac{1}{2} \|P\|_{2,I}^2. \quad (\text{A.90})$$

Thus

$$\|P\|_{2,I} \leq \sqrt{2} \|P\|_{\text{trap},I}. \quad (\text{A.91})$$

[Continuous RMS transfer before input rounding] Under equation A.84,

$$\|g\|_{2,\Omega} \leq \sqrt{\frac{2M}{M-1}} \|g\|_{2,X} + \frac{2(1+\sqrt{2})(B+T)}{\varrho-1} \varrho^{-p}. \quad (\text{A.92})$$

Proof. On each cell, equation A.88 and equation A.91 give

$$\|g\|_{2,I} \leq \sqrt{2} \|g\|_{\text{trap},I} + (1+\sqrt{2})e_p.$$

Combine cells by Minkowski's inequality. The global trapezoidal norm satisfies

$$\|g\|_{\text{trap}}^2 = \frac{1}{M-1} \left[\frac{|g(z_0)|^2 + |g(z_{M-1})|^2}{2} + \sum_{i=1}^{M-2} |g(z_i)|^2 \right] \leq \frac{M}{M-1} \|g\|_{2,X}^2.$$

Substitute equation A.88. □

A.8.1 INPUT ROUNDING AND COMPUTED EVALUATION

Let $Q(x)$ be the nearest working-format number to x , clamped to $[-1, 1]$, and let

$$\xi \geq \sup_{x \in \Omega} |Q(x) - x|. \quad (\text{A.93})$$

On a degree- p polynomial, standard Legendre estimates give

$$\|P\|_{\infty,I} \leq (p+1) \|P\|_{2,I}, \quad \|P'\|_{\infty,I} \leq \frac{2p^2(p+1)}{h} \|P\|_{2,I}. \quad (\text{A.94})$$

Define

$$\chi = \frac{2p^2(p+1)\xi}{h}, \quad C_X = (1+\chi) \sqrt{\frac{2M}{M-1}}, \quad C_A = \frac{2[1+\sqrt{2}(1+\chi)]}{\varrho-1} \varrho^{-p}. \quad (\text{A.95})$$

The same local argument as above yields

$$\|g \circ Q\|_{2,\Omega} \leq C_X \|g\|_{2,X} + C_A(B+T). \quad (\text{A.96})$$

Cauchy's estimate on the target gives $\|f'\|_{L^\infty(\Omega)} \leq B/\delta$, hence

$$\|f - f \circ Q\|_{2,\Omega} \leq \frac{B\xi}{\delta}. \quad (\text{A.97})$$

Finally evaluate the stored network at $Q(x)$ using fused affine arguments, the validated tanh routine, rounded products, and pairwise summation including the bias. By equation A.74,

$$|\hat{q}_W(x) - q(Q(x))| \leq (\epsilon_\phi + \Gamma_n^{\text{pw}})T + \beta_n^{\text{pw}}. \quad (\text{A.98})$$

Theorem A.3 (Computed continuous-RMS guarantee). *Under the analytic, sampling, and numerical conditions above,*

$$\|\hat{c}_W\|_1 \leq T, \quad (\text{A.99})$$

and

$$\|f - \hat{q}_W\|_{2,\Omega} \leq C_X R_X + C_A(B + T) + \frac{B\xi}{\delta} + (\epsilon_\phi + \Gamma_n^{\text{pw}})T + \beta_n^{\text{pw}}. \quad (\text{A.100})$$

Proof. The coefficient statement is equation A.83. For the function error, decompose

$$f - \hat{q}_W = (f - f \circ Q) + (g \circ Q) + (q \circ Q - \hat{q}_W).$$

Apply equation A.97, equation A.96, equation A.83, and equation A.98. $\square$

Roadmap checkpoint. R3 and R4 are complete. The sampled recovered readout has been transferred to continuous RMS, and the floating-point contributions appear as explicit allowances. The transfer combines the local analytic residual bound with standard Bernstein-ellipse approximation, polynomial inverse inequalities, pairwise summation, and backward-stable SVD.

A.9 THEOREM ASSEMBLY AND GEOMETRIC-TO-ALLOWANCE RATE

Substitute equation A.60 into equation A.82 and then into Theorem A.3. Define

$$A_{\text{rep}} = \frac{128\pi}{\lambda\delta}. \quad (\text{A.101})$$

Then

$$\begin{aligned} \|f - \hat{q}_W\|_{2,\Omega} &\leq 2C_X A_{\text{rep}} B e^{-c_{\text{app}} W} + \frac{2C_X A_{\text{rep}} B}{W} e^{-\pi^2/\lambda} \\ &\quad + C_X(4a_W S + 3b_W + \rho_W + e_{c,W}) + C_A(B + T) + \frac{B\xi}{\delta} + (\epsilon_\phi + \Gamma_n^{\text{pw}})T + \beta_n^{\text{pw}}. \end{aligned} \quad (\text{A.102})$$

Thus one valid choice in Theorem A.1 is

$$C_f = 2C_X A_{\text{rep}} B, \quad (\text{A.103})$$

with

$$\epsilon_{\text{rec},W} = C_X(4a_W S + 3b_W + \rho_W + e_{c,W}) + C_A(B + T) + \frac{B\xi}{\delta} + (\epsilon_\phi + \Gamma_n^{\text{pw}})T + \beta_n^{\text{pw}}. \quad (\text{A.104})$$

Likewise, since $2LS \leq 2BK_{\text{ref}}$, one may take

$$C_{\text{out}} = 2BK_{\text{ref}}, \quad \epsilon_{\text{coeff},W} = e_{c,W}. \quad (\text{A.105})$$

This proves Theorem A.1.

A.9.1 GEOMETRIC REFINEMENT ABOVE THE NONGEOMETRIC ALLOWANCE

For a fixed design, write the total nongeometric contribution as

$$\tau_W := \frac{2C_X A_{\text{rep}} B}{W} e^{-\pi^2/\lambda} + \epsilon_{\text{rec}, W}. \quad (\text{A.106})$$

Corollary A.3.1 (Logarithmic sufficient width above a fixed allowance). *Fix $\lambda, \delta, \Delta, a_H$ and a certified range of admissible widths $W \geq W_0$. Suppose*

$$\tau_W \leq \tau \quad \text{throughout that range.} \quad (\text{A.107})$$

Then

$$\|f - \hat{q}_W\|_{2, \Omega} \leq 2C_X A_{\text{rep}} B e^{-c_{\text{app}} W} + \tau. \quad (\text{A.108})$$

For any target tolerance $\epsilon > \tau$, it is sufficient that

$$W \geq \frac{1}{c_{\text{app}}} \log \left(\frac{2C_X A_{\text{rep}} B}{\epsilon - \tau} \right), \quad (\text{A.109})$$

provided the selected width lies in the certified range.

Proof. Equation A.108 follows by isolating the geometric term in equation A.102 and using equation A.107. Solving $2C_X A_{\text{rep}} B e^{-c_{\text{app}} W} \leq \epsilon - \tau$ gives equation A.109. $\square$

Scope of the logarithmic statement. Corollary A.3.1 holds for fixed λ , sampling budget, and arithmetic format over the certified width range. Its rate constant is $c_{\text{app}} = \frac{1}{8} \min\{\pi\delta, \lambda a_H^2\}$, while the readout constant also depends on λ . These dependencies are kept explicit when the design parameters are varied.